Lecture on

Waves & Oscillations

Classification of Motions

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Classification of Motions

Waves & Oscillations

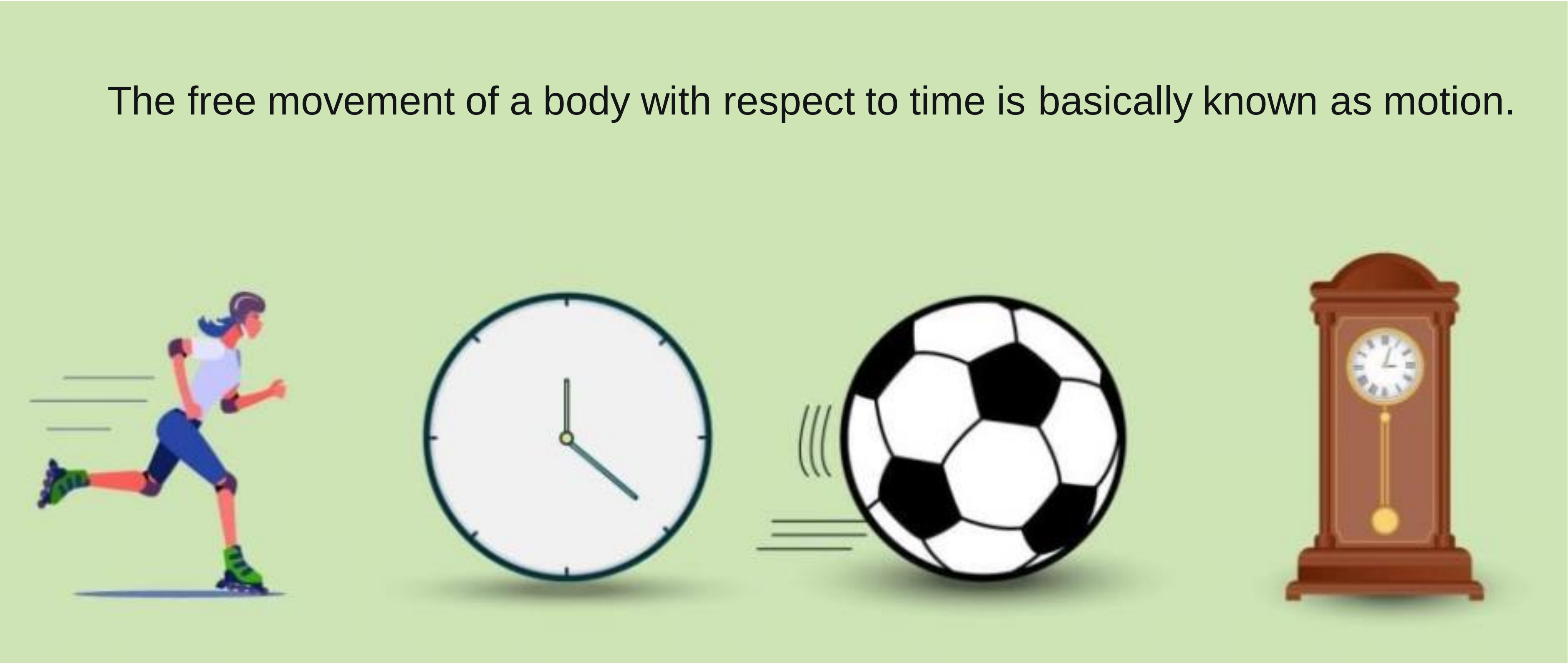
Where there is Wave there is Oscillation

Where there is Oscillation there is Wave

Motion is the common factor of Waves & Oscillations.

Classification of Motions

There are various types of motions that are happening around us and they can be basically distinguished on the basis of: Time, Speed, Distance, Dimension, Path, etc.



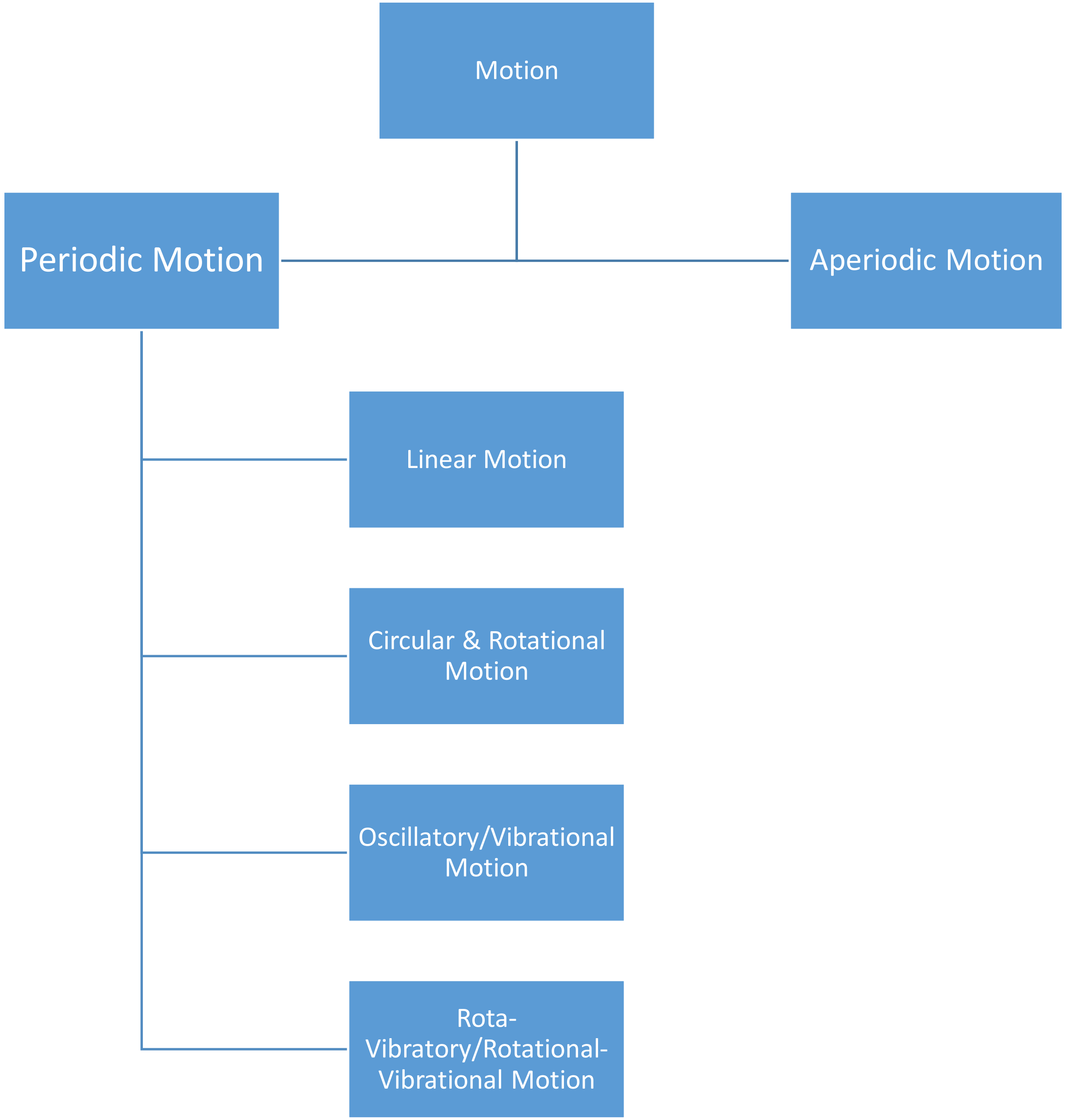
Example- the motion of fan, pendulum motion, the dust falling from the carpet, moving a train, piston movement, the water that flows from the tap, door lock rotation, sewing machine movement, a ball rolling around, motion of a clock, motion of a man during racing, a moving car, metronome, motion of universe, etc.

Classification of Motions

Suppose you are sitting on your sofa watching TV. Now, think about that, you are in motion or you are at rest?

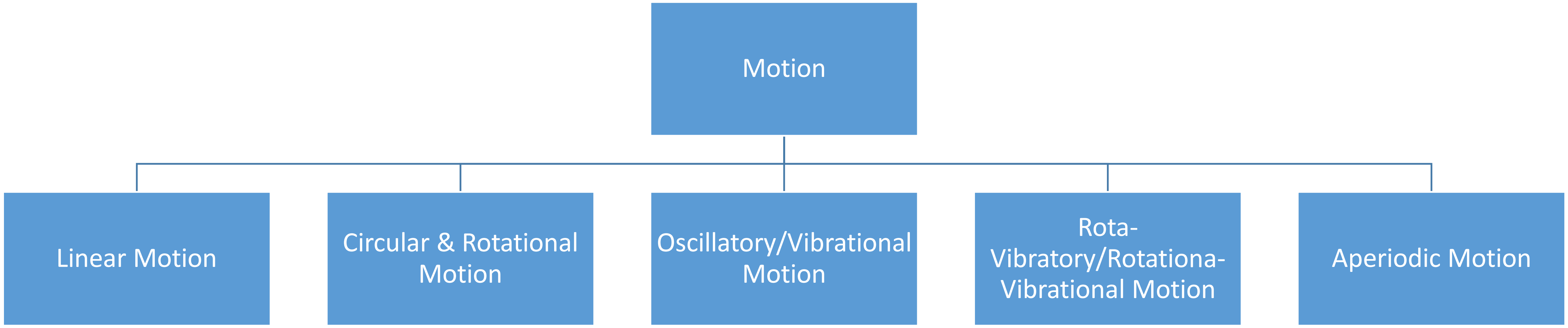
Before your answer, You must gather thorough proper knowledge about various **types of motion.**

Classification of Motions



Classification of Motions

In a gross below are the total types of motion:



Classification of Motions

Periodic Motion:

When a particle moves from one place to another at a set interval of time, then that motion is called Periodic Motion.

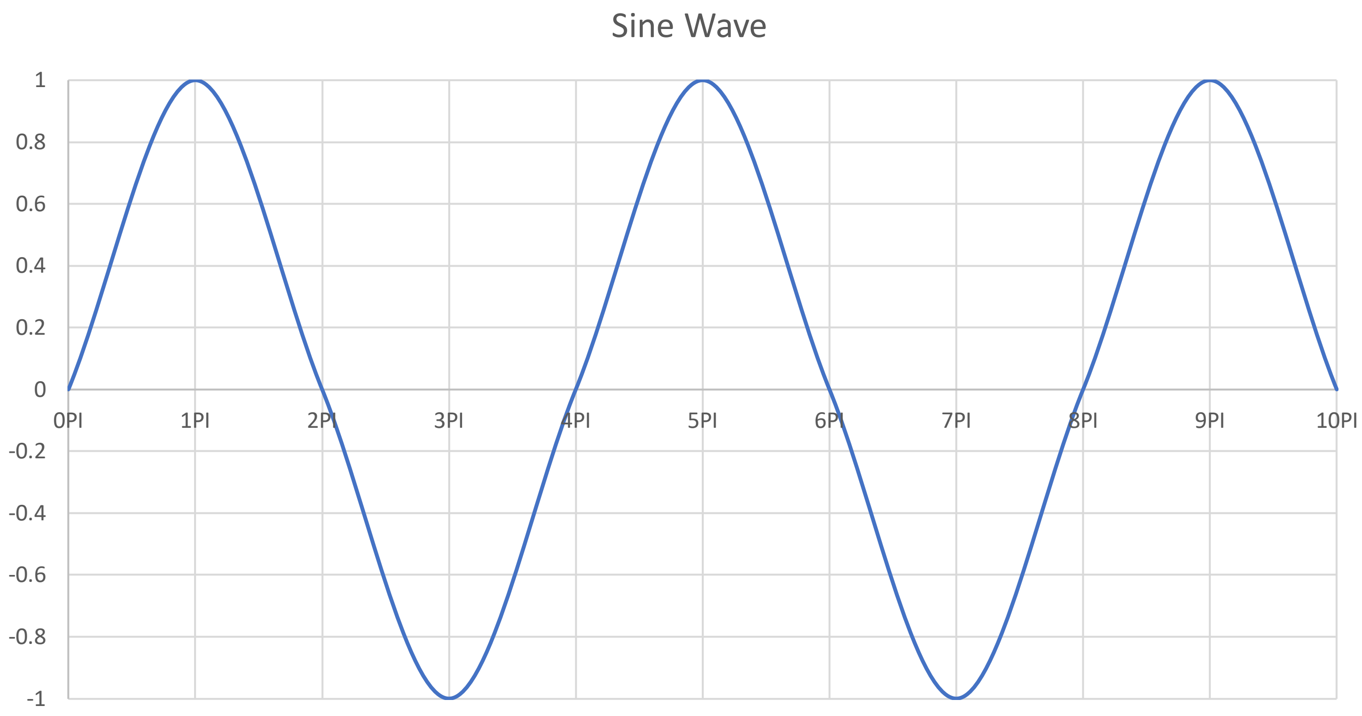


Fig: Periodic Wave

Time Period is constant? How?

Classification of Motions

Types:

1. ***Linear/Translational Motion:*** *When a particle is moving in a line, the motion is called linear motion or translational motion.*

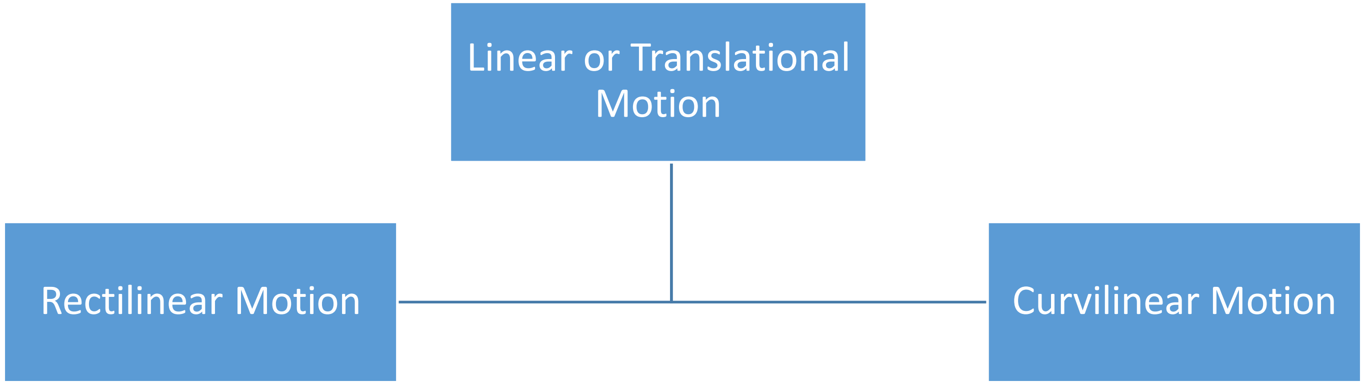
When a body is shifted or moved from one point to another point, then the body said to be experienced translational motion. More precisely, Translational motion is the motion in which all points of a moving body move uniformly in the same line or direction. ***Linear motion or Translatory motion/translational motion occurs when all points in a body move the same distance in the same amount of time***.

Example: A car moving from left to right, motion of a train, man walking on road, a bullet thrown from a gun at an angle, etc.

Fig: A man walking on the road

Classification of Motions

***Types**: 1. Linear/Translational Motion:*



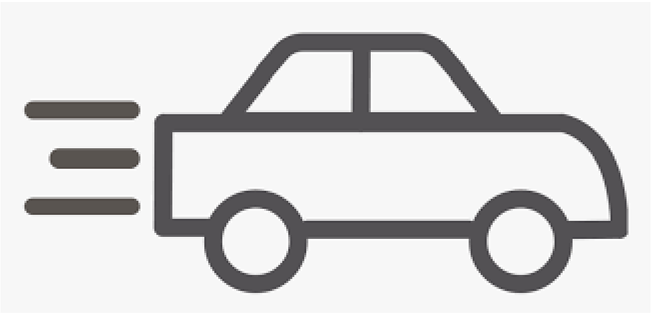


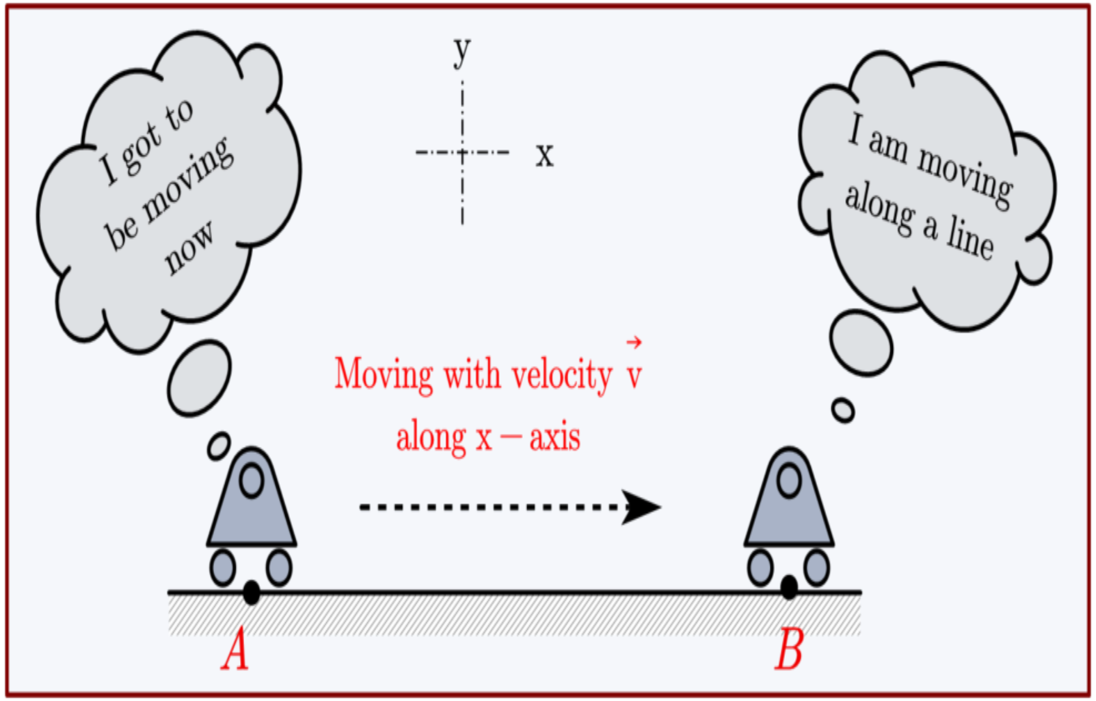
Fig: Car moving on a straight path

Classification of Motions

Types*: *1.* *Linear/Translational Motion: There are two types of linear motion. (i) Rectilinear Motion and (ii) Curvilinear Motion.

(i) Rectilinear Motion: *When an object in translational motion moves in a straight line, it is said to be in rectilinear motion.* In such type of motion, the ***Body is moving in a straight line or straight path without rotation***. Simply, The path of the motion is a straight line.

Example: Car moving on a straight path, A bullet fired from the gun, a train moving on a straight track, etc.

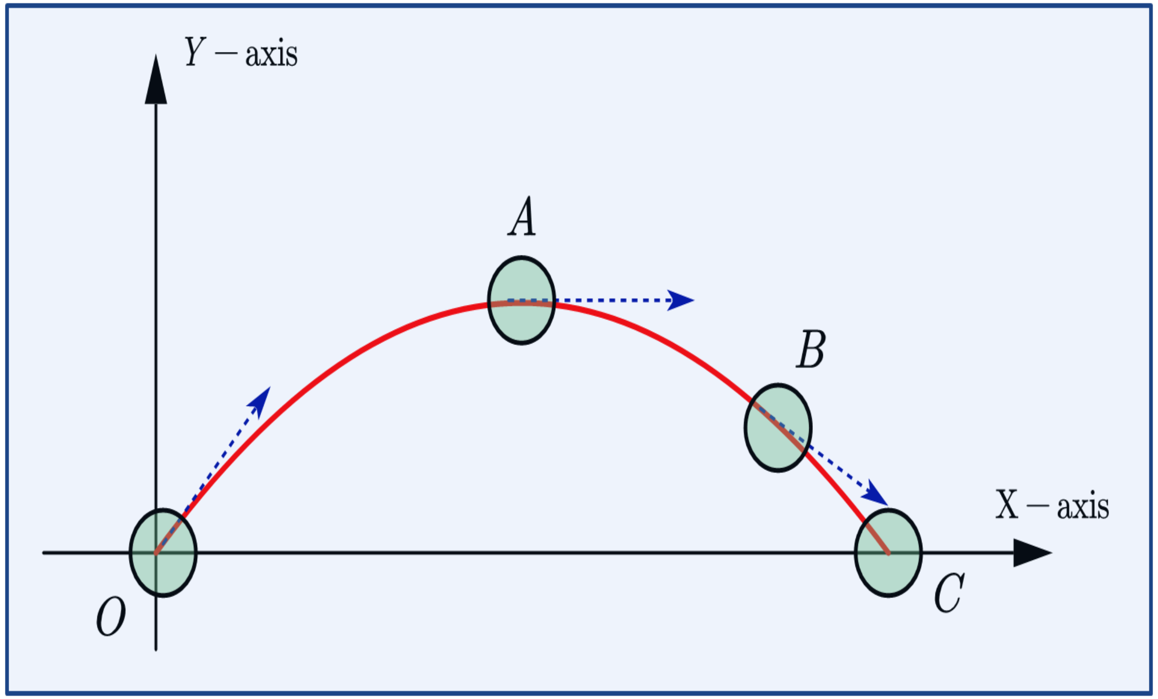


Classification of Motions

Types*: *1. Linear/Translational Motion: There are two types of linear motion. (i) Rectilinear Motion and (ii) Curvilinear Motion.

(ii) Curvilinear Motion: *When an object in translational motion moves along a curved path, it is said to be in curvilinear motion.* In such type of motion, the ***Body is moving in a curved path without rotation***. Simply, The path of the motion is curved.

Example: Car moving on a curved path, A stone thrown up in the air, projectile motion, etc.



Classification of Motions

2. Circular & Rotational Motion: *When a particle is moving in a circular path, the motion is said to be Circular/Rotational Motion or Circular & Rotational motion*. Example: Motion of the hands of a clock, Motion of a wheel, etc.

Here, F_W = F_{Cf} means centrifugal and $F_Z = F_{Cp}$ means centripetal force

F_{Cp} is required for circular motion
F_{Cf} is also required for circular motion
And F_{Cp} & F_{Cf} both are required for circular and rotational motion

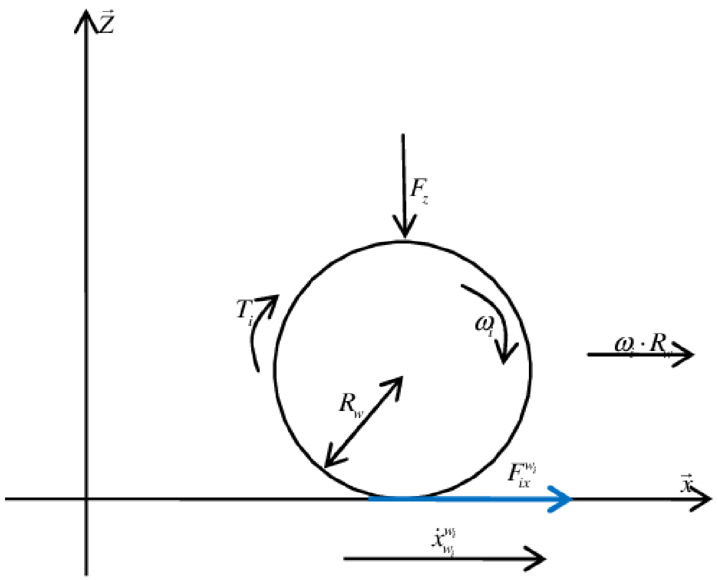


Fig: Motion of a wheel

$$F = m\omega^2 r$$

$$F_{Cf} = m\omega^2 r = ma$$

$$F_c = m a_c \qquad a_c = \frac{v^2}{r}$$

$$F_{Cp} = ma_c$$

Classification of Motions

Rotational Factor or Rotational Character:

- Moment of Inertia, $I = MR^2 = mr^2$
- Angular Momentum, $L = \sum r \times P$ $\quad \therefore \vec{L} = \vec{r} \times \vec{P}$
- Torque, $\vec{\tau} = \vec{r} \times \vec{F} = \mathrm{I}\alpha$
- Angular acceleration, $a = \alpha = \frac{d\omega}{dt} = \frac{\Delta\omega}{\Delta t} = \frac{a_T}{r}$
- Acceleration, $a_c = \frac{{v_T}^2}{r}$
- Angular velocity, $\omega = \frac{d\theta}{dt} = \frac{\Delta\theta}{\Delta t} = \frac{v_T}{r} = \frac{v}{r}$
- Axis of Rotation, $\Delta\theta = \theta_f - \theta_i$
- Angular Displacement, $\theta = \omega t$ or $d\theta$
- Rotational Kinetic energy, $K = \frac{1}{2} I\omega^2$
- Rotational Power, $P = \tau\omega = \tau\frac{d\theta}{dt}$
- Rotational Work done,
  $W = \int \tau d\phi = \int \tau d\theta = \tau\theta$
- Rotational force, $F = mr\alpha$

Example-2: A constant torque of 500 kN·m is applied to a wind turbine to keep it rotating at 6 rad/s. What is the power required to keep the turbine rotating?

3×10^6W

Soln: Try yourself

- Rotational Frequency, $\nu = \frac{\omega}{2\pi}$
- Rotation Period, T=ν^{-1}
- RPM
- Centripetal force, $F_c = F_{cp} = ma_c = m\frac{v^2}{r}$
- Centrifugal force, $F_{cf} = -ma_c = -m\omega^2 r$
- Angular Speed, $\omega = \frac{d\theta}{dt} = \frac{\Delta\theta}{\Delta t} = \frac{v_T}{r} = \frac{v}{r}$

Example-1: A boat engine operating at 9×10^4 W is running at 300 rev/min. What is the torque on the propeller shaft?

Soln:

$$300.0\,\mathrm{rev/min} = 31.4\,\mathrm{rad/s};$$

$$\tau = \frac{P}{\omega} = \frac{9.0 \times 10^4 \mathrm{N \cdot m/s}}{31.4\,\mathrm{rad/s}} = 2864.8\,\mathrm{N \cdot m}.$$

Classification of Motions

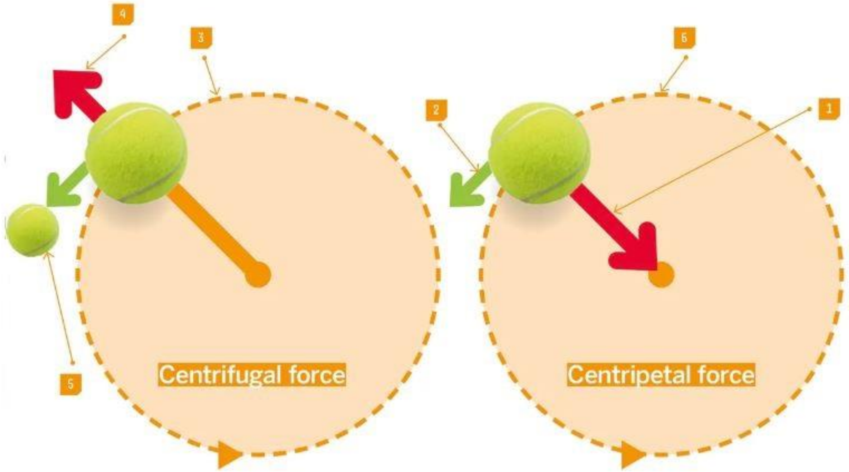
A centripetal force is the total radial force acting inwards on an object in a circular motion.
Example: Electrons orbit around the nucleus of the atom, Driving a vehicle on curves, An orbiting planet, A roller coaster ride, etc.

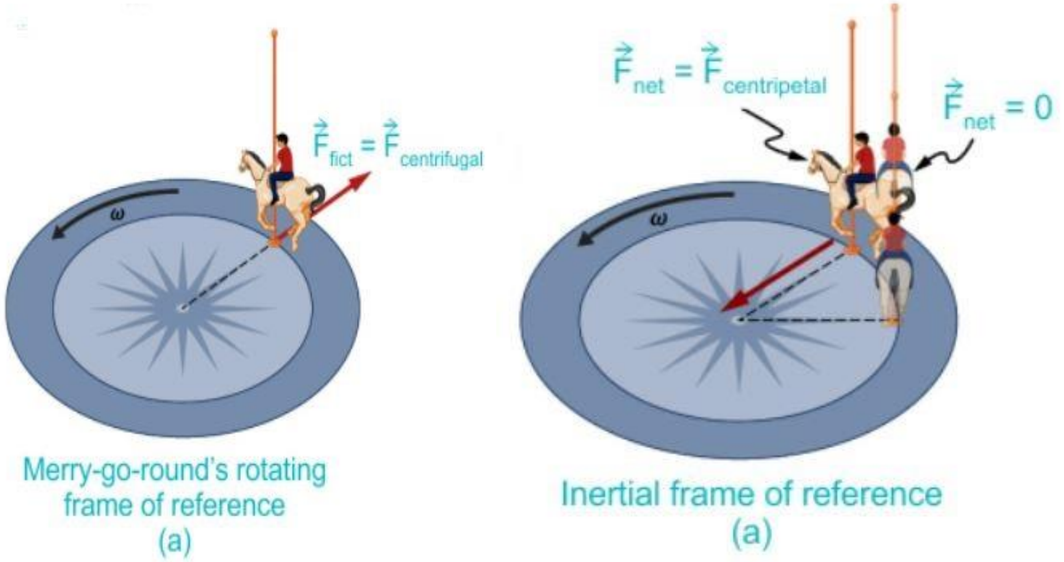
From an observer's reference frame, if we ignore gravity there is only one force acting on the ball on the string: the **tension force** from the string holding it. The **tension** provides the radial force necessary to keep it in a circular motion. The net radial force acting on an object that keeps it moving in a circle.

$$F_{Cp} = ma_c = m\frac{v^2}{r}$$

A centrifugal force is an apparent force directed radially outwards felt by an object in a rotating reference frame.
Example: The spinning of the washing machine, Banking of the roads, The spinning of the merry-go-round, centrifugal pump machines, etc.

We use a "fictitious force" which is the **centrifugal force**. The centrifugal force is directed radially outwards and has the same magnitude as the centripetal force. $F_{Cf} = m\omega^2 r = ma$





Classification of Motions

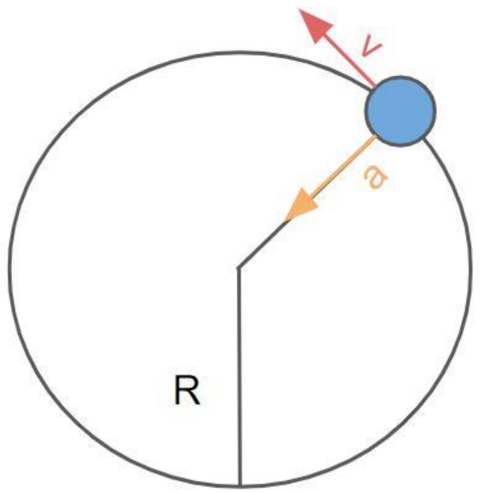
Difference Between Centripetal and Centrifugal Force

The difference between centripetal and centrifugal forces are mentioned as follows below:

Centripetal Force	Centrifugal Force
Its direction is towards the center of curvature.	Its direction is away from the center of curvature.
It is observed from the inertial frame of the reference.	It is observed from the non-inertial frame of reference.
Its force is towards the axis of rotation.	Its force is outward of the axis of rotation.
It is considered a real force.	It is considered a pseudo-force.

Classification of Motions

Uniform circular motion is the motion of an object moving with constant speed in a circle with a fixed radius.

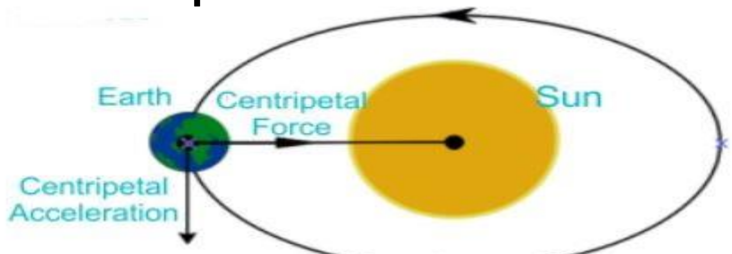


Let's consider a ball on a string moving in a circle, as shown below. If the ball is moving with a constant speed around the circle, it is experiencing **uniform circular motion**. **Example:** Any point on a propeller spinning at a constant rate, The second, minute, and hour hands of a watch, etc.

Centripetal acceleration is the radial acceleration of an object in a circular motion.

$$a_c = \frac{v^2}{r}$$

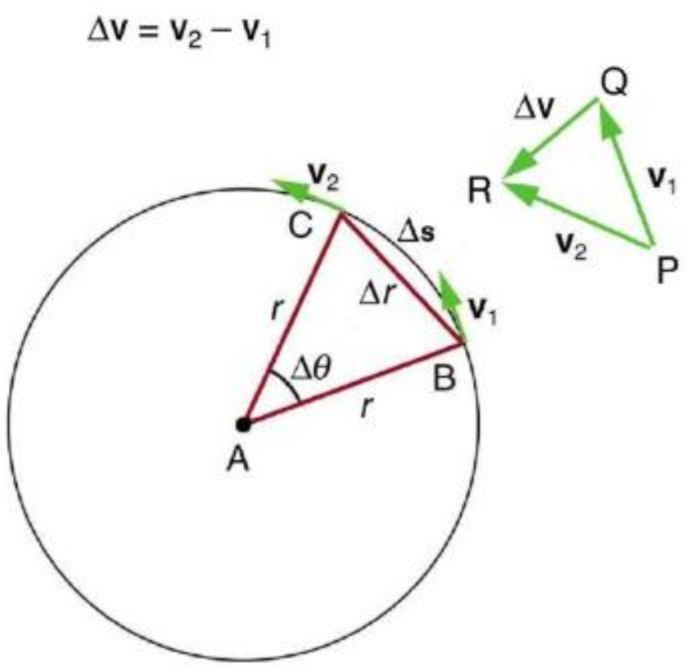
A ball on a string in a uniform circular motion. As shown in the image above, ***the velocity of the ball is always in a direction tangent to the circle and thus the direction of the velocity vector is constantly changing.*** This causes the ball to experience acceleration perpendicular to the velocity vector, and thus the acceleration vector is always pointing to the center of the circle, as shown in the image. This radial acceleration is known as the **centripetal acceleration**.



Centripetal Force Illustration

Classification of Motions

 # Centripetal acceleration:



Triangle formed by the velocity vectors and the one formed by the radii *r* and *Δs* are similar. Both the triangles ABC and PQR are isosceles triangles (two equal sides). The two equal sides of the velocity vector triangle are the speeds $v_1 = v_2 = v$. Using the properties of two similar triangles, we obtain

$$\frac{\Delta v}{v} = \frac{\Delta s}{r}. \qquad \Delta v = \frac{v}{r}\Delta s$$

$$\frac{\Delta v}{\Delta t} = \frac{v}{r} \times \frac{\Delta s}{\Delta t}. \qquad \text{But, } \frac{\Delta v}{\Delta t} = a_c \quad \text{and} \quad \frac{\Delta s}{\Delta t} = v$$

$$a_c = \frac{v^2}{r}; \; a_c = r\omega^2$$

Tangential velocity, $v = \omega r$

Tangential acceleration, $\mathrm{a} = \frac{\Delta v}{\Delta t} = \mathrm{r} \propto$

Centripetal acceleration, $a_c = \frac{{v_T}^2}{r}$

Angular acceleration, $\propto = \frac{\Delta \omega}{\Delta t}$

Example-3: What is the magnitude of the centripetal acceleration of a car following a curve of radius 500 m at a speed of 25.0 m/s (about 90 km/h)? Compare the acceleration with that due to gravity for a fairly gentle curve of the car taken at highway speed.

Soln: $a_c = \frac{v^2}{r} = \frac{(25.0 m/s)^2}{500\, m} = 1.25\, m/s^2.$ and $a_c/g = (1.25\, m/s^2)/(9.80\, m/s^2) = 0.128.$ $a_c = 0.128g$

Classification of Motions

 # Centripetal acceleration:

Example-4: Calculate the centripetal acceleration of a point 7.50 cm from the axis of an **ultracentrifuge** spinning at 7.4×10^7 rev/min. Determine the ratio of this acceleration to that due to gravity.

Soln: Try yourself $a_c = 4.5 \times 10^6 \text{ m}/s^2$

Example-5: Calculate the angular speed of a 0.300 m radius car tire when the car travels at 15.0 m/s (about 54 km/h).

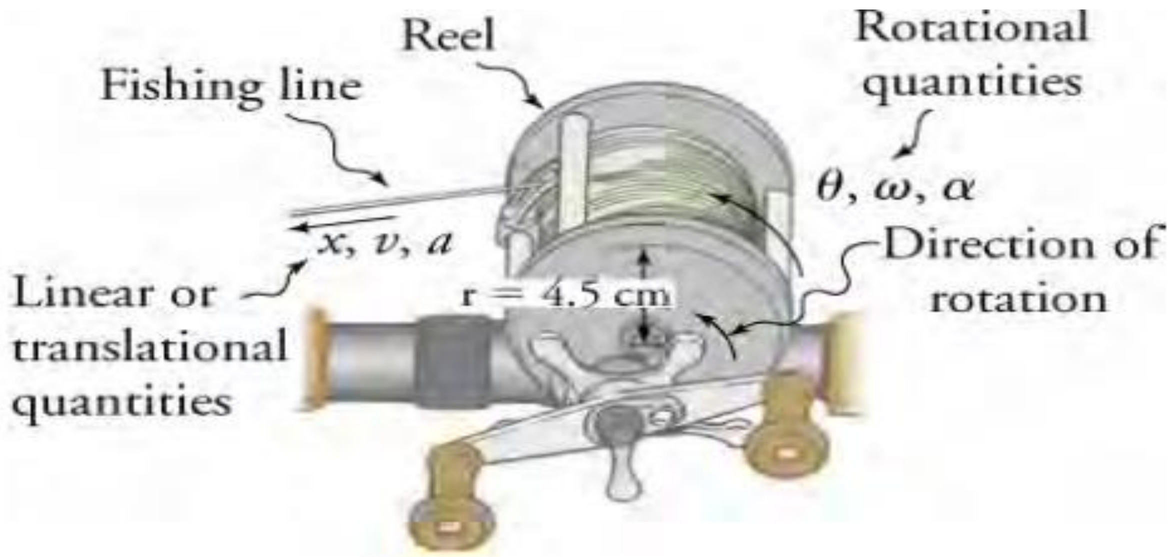
Soln: 50 rad/s

Example-6: Calculate the centripetal force exerted on a 900 kg car that rounds a 600-m-radius curve on horizontal ground at 25.0 m/s.

Soln: Try yourself 938 N

Example-7: A deep-sea fisherman uses a fishing rod with a reel of radius 4.50 cm. A big fish takes the bait and swims away from the boat, pulling the fishing line from his fishing reel. As the fishing line unwinds from the reel, the reel spins at an angular velocity of 220 rad/s. The fisherman applies a brake to the spinning reel, creating an angular acceleration of −300 rad/s2. How long does it take the reel to come to a stop?

Classification of Motions



$$\omega_o = 220 \frac{rad}{s}$$
$$\omega = 0$$
$$\alpha = -300\, rad/s^2$$

Solution

The equation to use is $\boldsymbol{\omega} = \boldsymbol{\omega}_0 + \boldsymbol{\alpha} t$.

We solve the equation algebraically for t, and then insert the known values

$$\begin{aligned} t &= \frac{\omega - \omega_0}{\alpha} \\ &= \frac{0-220 \text{ rad/s}}{-300 \text{rad/s}^2} \\ &= 0.733 \text{ s} \end{aligned}$$

Classification of Motions

2. Circular & Rotational Motion:

Circular and Rotational Motion

- Circular Motion
- Rotational Motion

Fig: Circular motion

Classification of Motions

2. Circular & Rotational Motion: The circular & rotational motion are classified mainly in two branches. (i) Circular Motion and (ii) Rotational Motion.

(i) Circular Motion: *The body rotates about a fixed axes which lies outside the body.*
When an object is constantly moving in circular a path it is called circular motion. **In a circular motion, the speed of the object should be constant.**

Example: Hands of a working clock, moon moves around earth, etc.

Average distance=384,400 km (238,900 mi) from Earth's centre

Time Period=27.3 days

Mean orbital speed of the Moon around the barycentre=1.022 km/s (2288 miles per hour≈2,290 mph)

Q: How can you find out the velocity of the Moon and its angular speed?

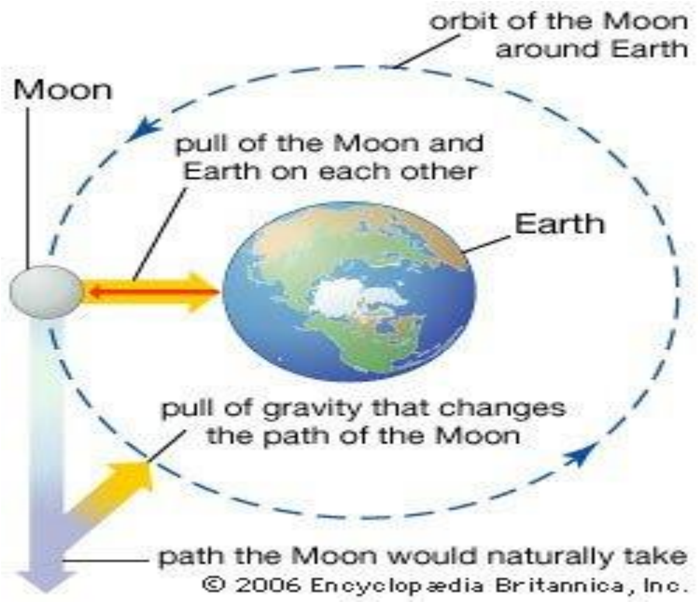
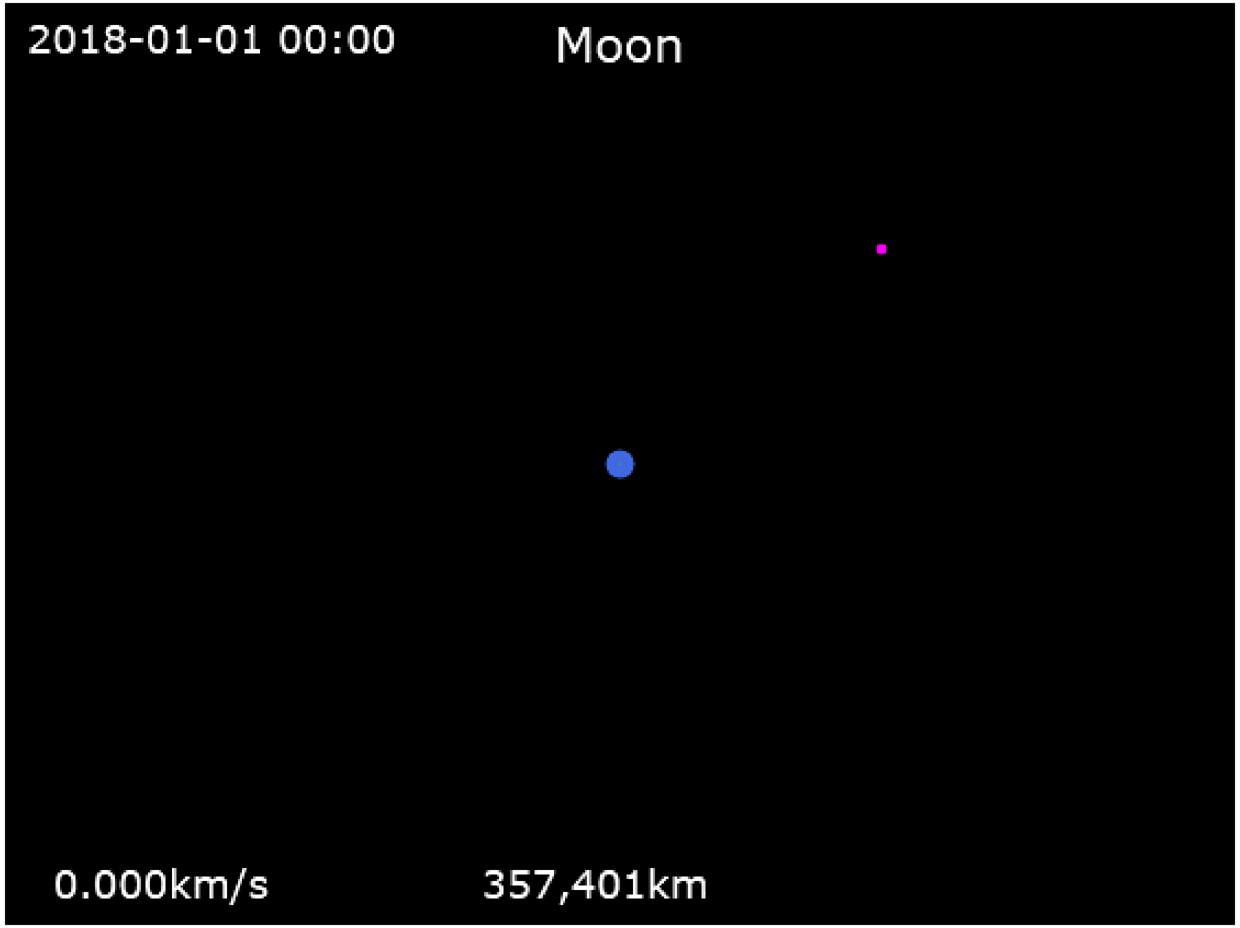


Fig: Moon moves around earth

Classification of Motions

2. Circular & Rotational Motion: (i) Circular Motion



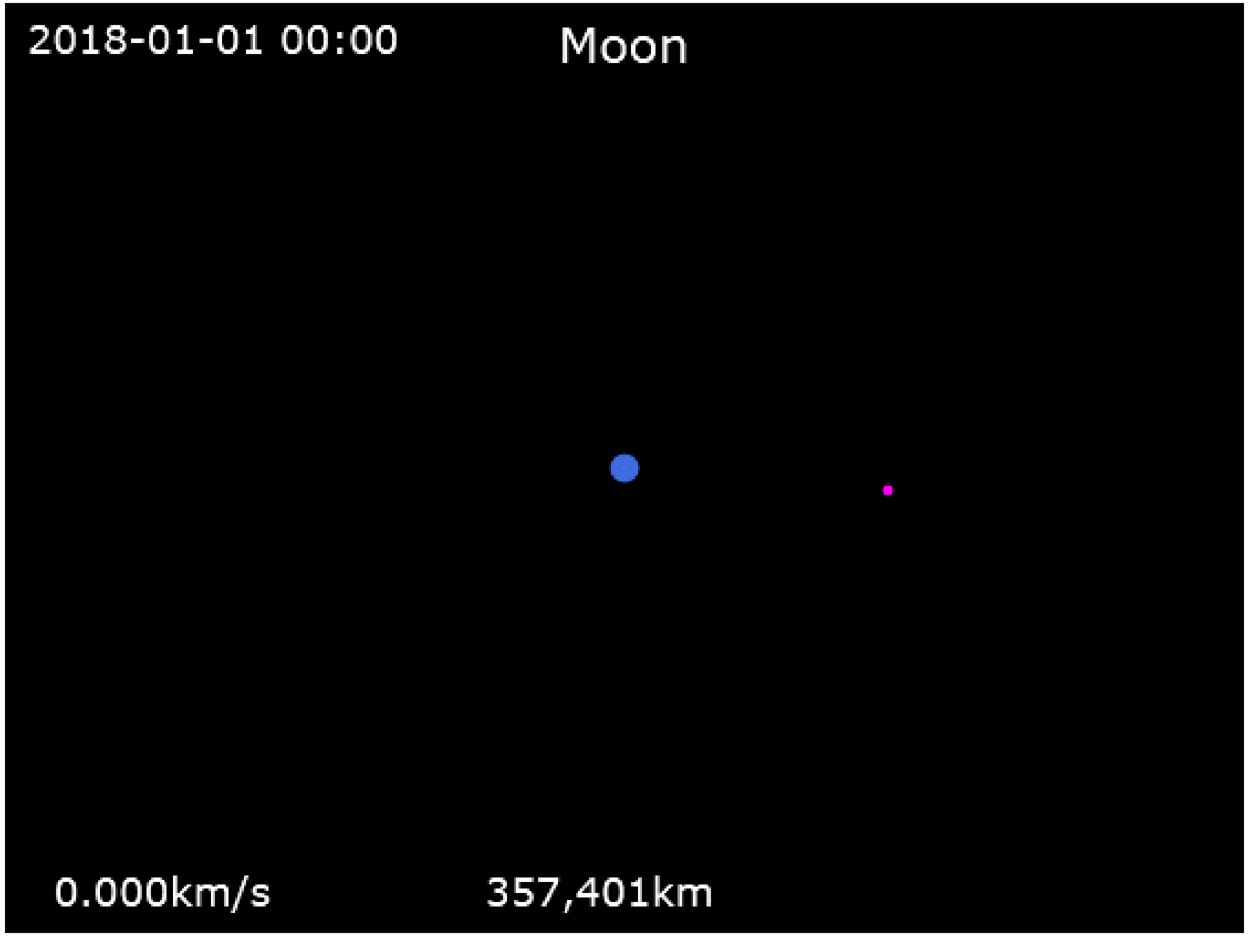


Fig: Moon orbits around Earth

Classification of Motions

The Moon's orbital plane is inclined by about 5.1° with respect to the ecliptic plane, whereas Earth's equatorial plane is tilted by about 23.4° with respect to the ecliptic plane.

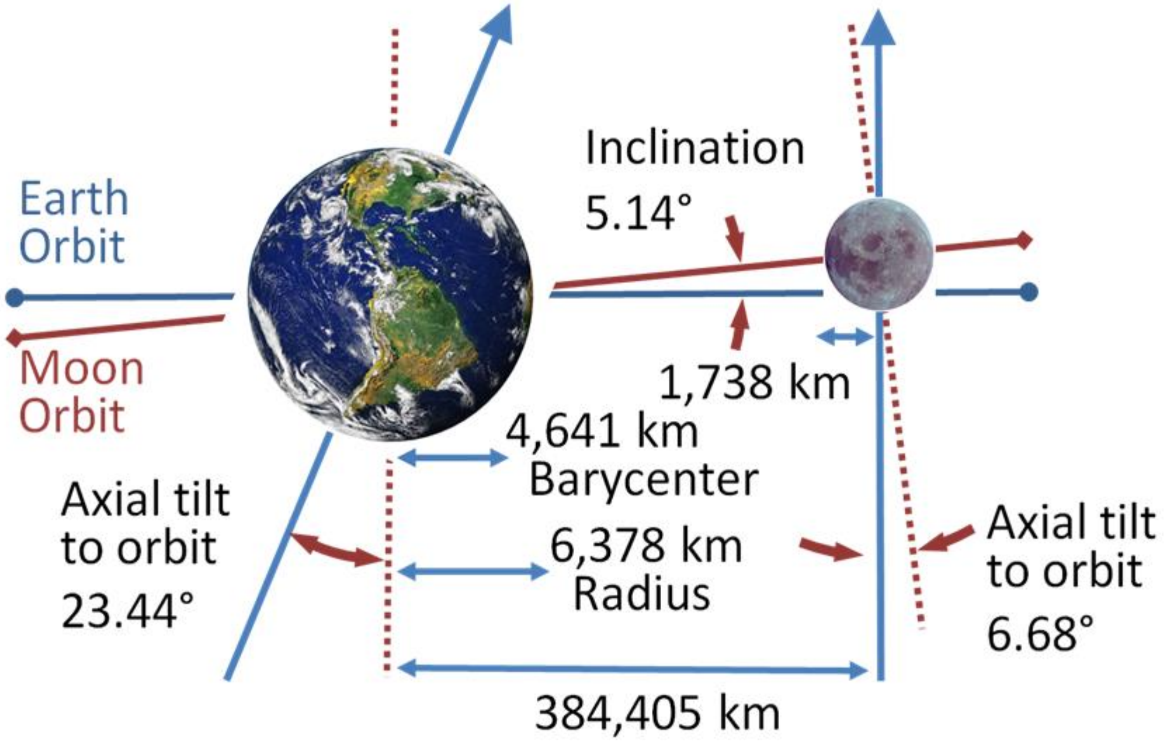


Fig: Orbital inclination-the Moon's orbit is inclined by 5.14° to the ecliptic

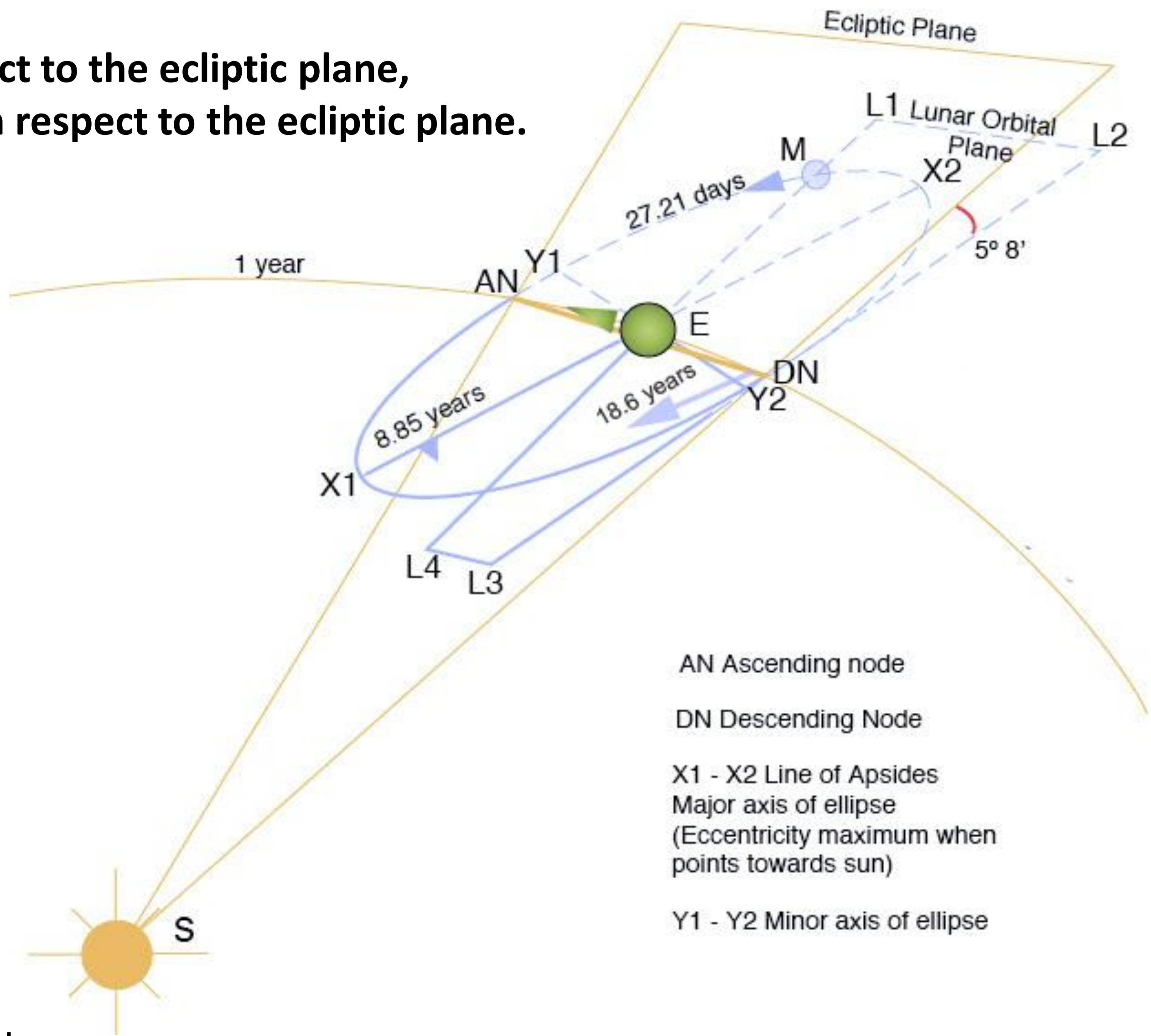


Fig: Earth's lunar orbit perturbations

Classification of Motions

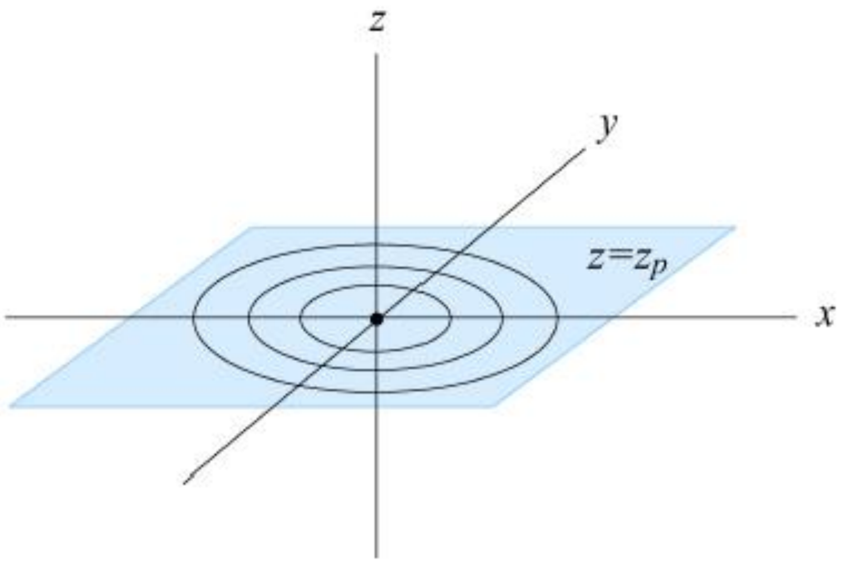


Figure 6.1 level surfaces for the coordinate r

The coordinates (r,θ) in the plane $z = z_p$ are called ***polar coordinates***.

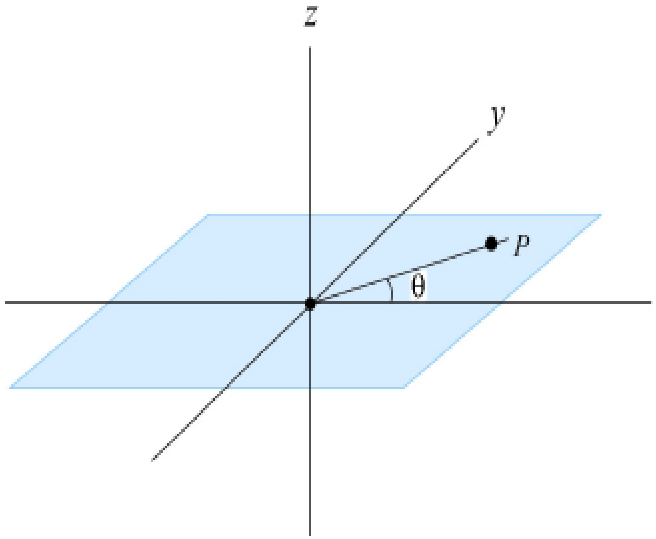


Figure 6.2 the angle coordinate

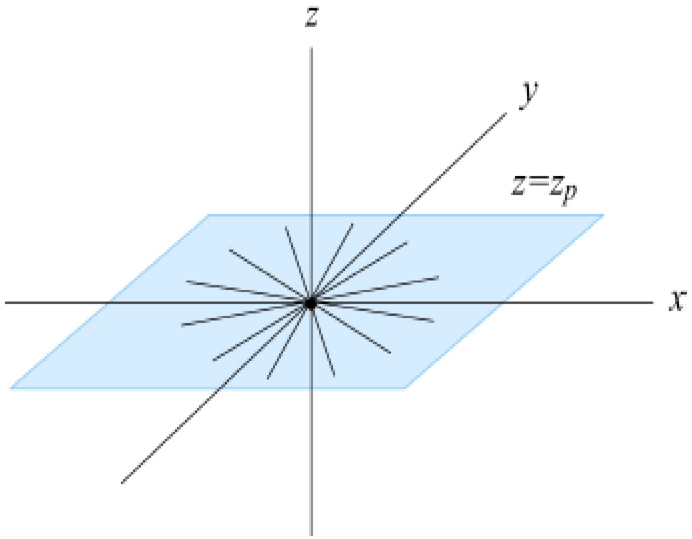


Figure 6.3 Level surfaces for the angle coordinate

Cylindrical Coordinate System

We choose two unit vectors in the plane at the point P as follows. We choose ˆ**r** to point in the direction of increasing r , radially away from the z-axis. We choose ˆθ to point in the direction of increasing θ . This unit vector points in the counterclockwise direction, tangent to the circle. Our complete coordinate system is shown in Figure 6.4. This coordinate system is called a 'cylindrical coordinate system'. Essentially we have chosen two directions, radial and tangential in the plane and a perpendicular direction to the plane. If we are given polar coordinates (r,θ) of a point in the plane, the Cartesian coordinates (x, y) can be determined from the coordinate transformations

$$x = r\cos\theta\ ,\quad y = r\sin\theta$$

Classification of Motions

Conversely, if we are given the Cartesian coordinates (*x*, *y*) , the polar coordinates (*r*,θ) can be determined from the coordinate transformations

$$\boldsymbol{r} = \sqrt{\boldsymbol{x}^2 + \boldsymbol{y}^2}\ , \quad \boldsymbol{\theta} = \mathbf{tan}^{-1}(\frac{\boldsymbol{y}}{\boldsymbol{x}})$$

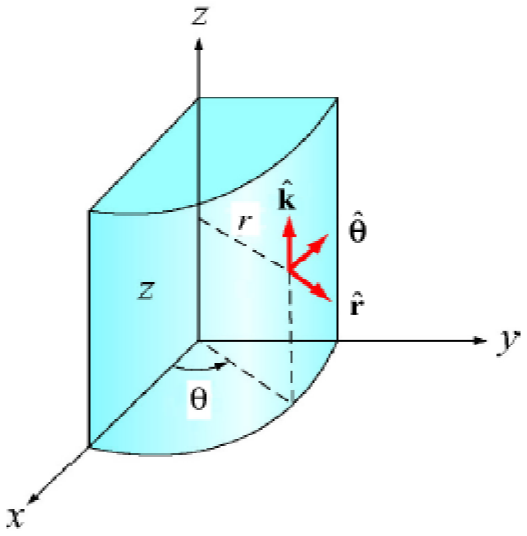


Figure 6.4 Cylindrical coordinates

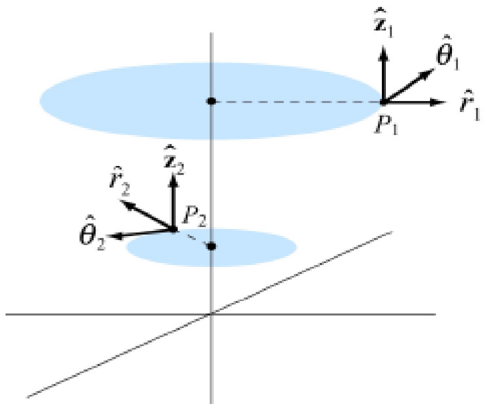


Figure 6.5 Unit vectors at two different points in polar coordinates.

The motion of an object moving around a circle at a constant speed can be modeled as follows. We impose a coordinate system with the origin at the center of the circle. The object's position can be specified by an angle measured counterclockwise from the positive x-axis. Then, $\boldsymbol{\theta}$=$\boldsymbol{\theta_0} + \boldsymbol{\omega t}$ $\quad \therefore \theta(t) = \omega t$ (if $\boldsymbol{\theta_0} = 0^0$)

where θ_0 is the angle of the initial position (i.e., the position at time t = 0), ω is the angular speed in radians per unit time, and the plus or minus, is take to be plus if the motion is counterclockwise, and minus if the motion is clockwise.

In xy-coordinate, the location of the object is then given by (x,y)= **(r cos(θ_0± ωt), r sin(θ_0± ωt))**

where r is the radius of the circle.

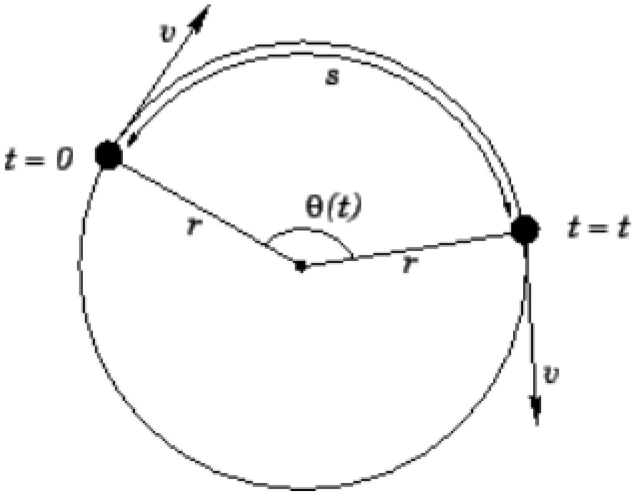


Figure : *Circular motion*

$$s = r\,\theta.$$

$$s = \frac{2\pi}{360^\circ}\, r\,\theta(^\circ).$$

$$v = r\,\omega.$$

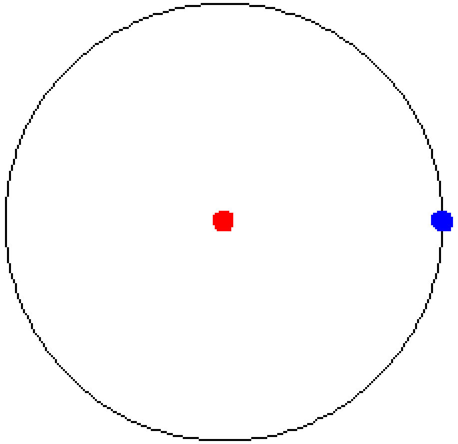
$$T = \frac{2\,\pi}{\omega}$$

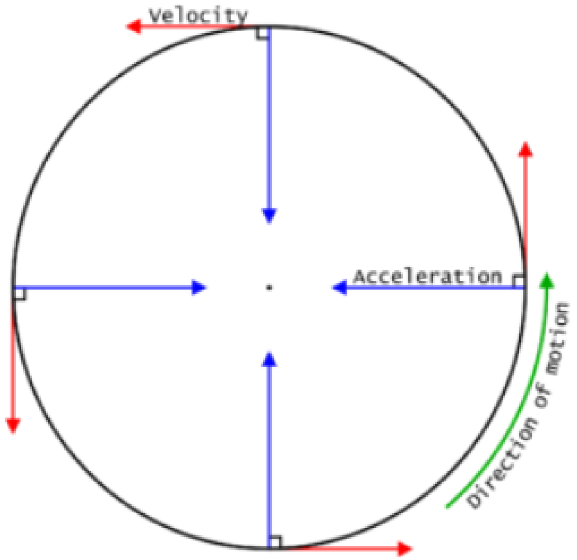
$$f = \frac{1}{T} = \frac{\omega}{2\,\pi}.$$

$$T = \frac{1}{f}$$

$$\omega = 2\pi\, f$$

$$x(t) = R\cos(\omega t)$$
$$y(t) = R\sin(\omega t).$$





Example-8: Suppose that an object executes uniform circular motion, radius r=1.2 m at a frequency of f =50 Hz. Calculate the time period, angular frequency and tangential velocity.

Soln: Try yourself

T=0.02 s, $\omega = 314.16\frac{rad}{s}$, $V_T = 376.992\ m/s$

Classification of Motions

2. Circular & Rotational Motion: The circular & rotational motion are classified mainly in two branches. (i) Circular Motion and (ii) Rotational Motion.

(ii) Rotational Motion: ***When an object moves about an axis and different parts of it move by different distances in a given interval of time, it is said to be in rotational motion***. Usually, Rotational motion deals with rigid bodies.

All the particles do not have the same displacement, velocity, and acceleration. So that, when an object moves along its axis, all the parts of it move for a different distance in a given period of time which means when an object is under rotational motion all of its parts will move different distances in the same interval of time. It spins around an internal axis. *Simply,* ***Rotational motion is the motion that occurs when a body rotates on its own axis***. In rotational motion, the ***Axes of rotation of a body lies inside the body.***

Example: Satellite Motion, skater rotating on an ice rink, drone motion, etc.

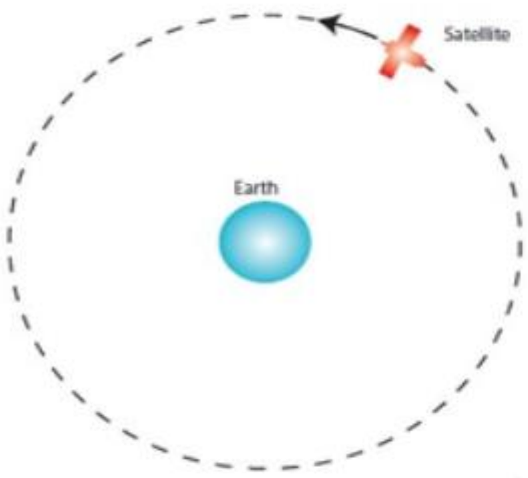


Fig: satellite motion

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion.

Satellite Motion

Satellite motion is physics surrounding the orbit of a small masses around larger masses. It is assumed that satellites have circular orbits.

Satellites can be either natural (planets, moons, etc.) or artificial (objects sent up into orbit by humans).

Some artificial satellites are in a geostationary orbit; that is that they stay above the same fixed location on Earth's surface throughout their orbit.

In order for this to occur these satellites must:

→orbit directly above the equator and in the same direction as Earth's rotation

→have an orbital period equal to Earth's rotational period – 24 hours.

Orbital motion

Orbital motion is motion of a body around another body, moving in a stable path, due to its gravitational attraction.

Consider firing a cannon at an angle from Earth at ever-increasing speeds yields one of three possibilities:

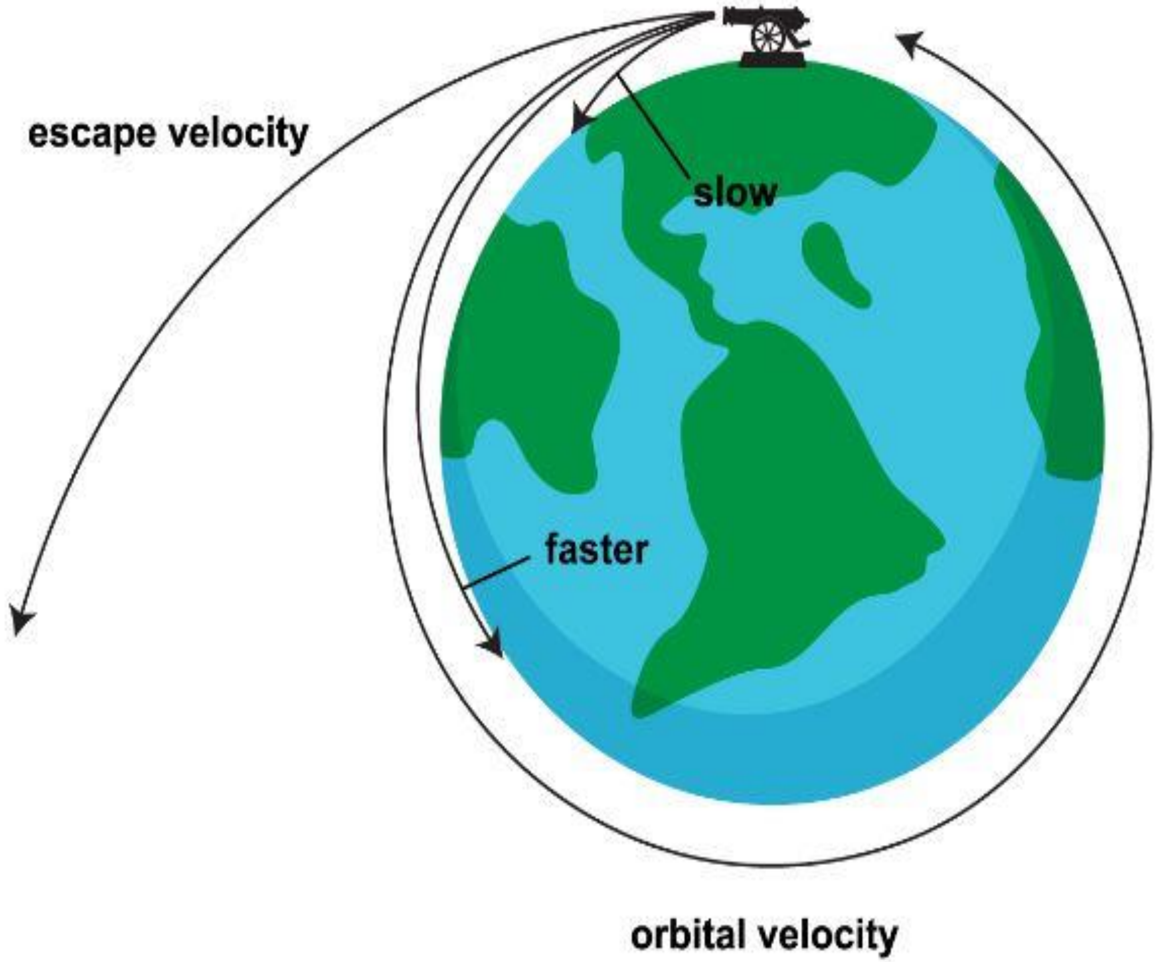
→The speed of the projectile is too slow, and the projectile hits the ground

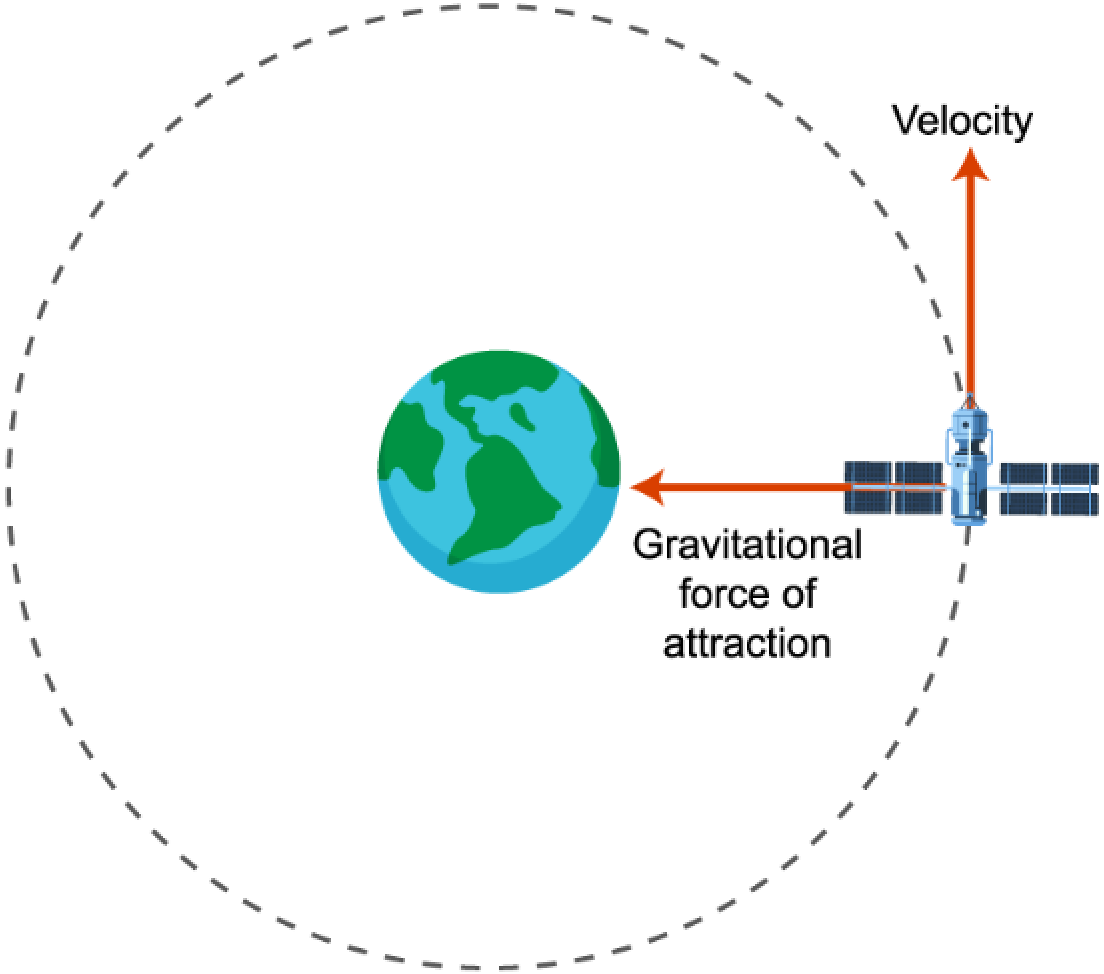
→The speed of the projectile it too fast and the projectile is sent into space

→The speed of the projectile is just right and the projectile enters a stable orbit around the Earth. In this case the projectile falls towards the Earth without ever getting closer to it, due to the curvature of the Earth.

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion





Circular orbital path

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion

Any satellite orbiting a central mass obeys equations of circular motion, as we assume its orbit is circular:

$$v = \frac{2\pi r}{T}$$

$$v = \sqrt{\frac{GM}{r}}$$

Where

v is the orbital velocity (m/s)

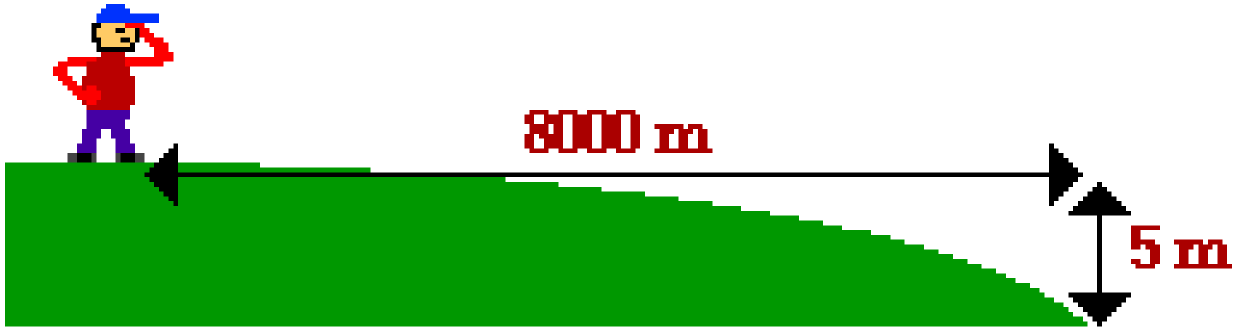
r is the orbital radius (m)

T is the orbital period (s)

G is the gravitational constant $(6.67 \times 10^{-11} Nm^2/kg^2)$

M is the mass of the central mass being orbited (kg)

To avoid hitting the Earth, an orbiting projectile must be launched with a horizontal speed of 8000 m/s. When launched at this speed, the projectile will fall towards the Earth with a trajectory which matches the curvature of the Earth. As such, the projectile will fall around the Earth, always accelerating towards it under the influence of gravity, yet never colliding into it since the Earth is constantly curving at the same rate. Such a projectile is an orbiting satellite.



Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion.

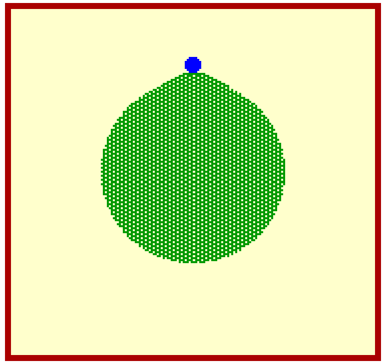


Fig-1: Launch Speed very very greater than 8000 m/s
Projectile escapes from orbits Earth

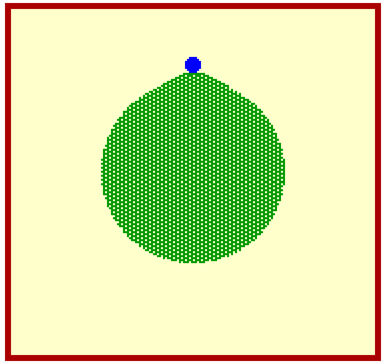


Fig-2: Launch Speed very very less than 8000 m/s
Projectile falls to Earth

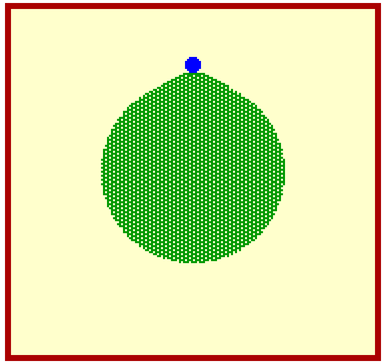


Fig-3: Launch Speed less than 8000 m/s
Projectile falls to Earth

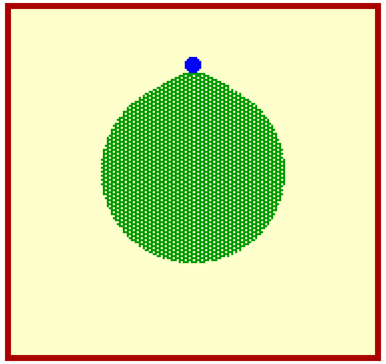


Fig-4: Launch Speed equal to 8000 m/s
Projectile orbits Earth - Circular Path

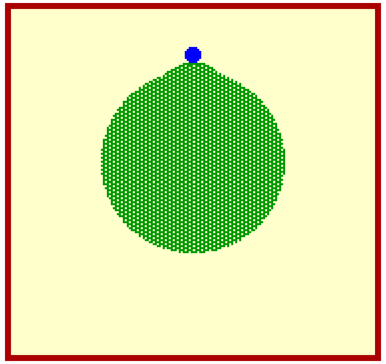


Fig-5: Launch Speed greater than 8000 m/s
Projectile orbits Earth - Elliptical Path

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion.

***Mathematics of Satellite Motion

Orbital Speed Equation

Consider a satellite with mass M_{sat} orbiting a central body with a mass of mass $M_{Central}$. The central body could be a planet, the sun or some other large mass capable of causing sufficient acceleration on a less massive nearby object.

The net centripetal force acting upon this orbiting satellite is given by the relationship

$$F_{net} = (M_{sat} \bullet v^2) / R$$

This net centripetal force is the result of the gravitational force that attracts the satellite towards the central body and can be represented as

$$F_{grav} = (G \bullet M_{sat} \bullet M_{Central}) / R^2$$

Since $F_{grav} = F_{net}$, the above expressions for centripetal force and gravitational force can be set equal to each other. Thus,

$$(M_{sat} \bullet v^2) / R = (G \bullet M_{sat} \bullet M_{Central}) / R^2$$

Observe that the mass of the satellite is present on both sides of the equation; thus it can be canceled by dividing through by $\mathbf{M_{sat}}$. Then both sides of the equation can be multiplied by **R**, leaving the following equation

$$v^2 = (G \bullet M_{Central}) / R$$

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion.

Orbital Speed Equation

Taking the square root of each side, leaves the following equation for the velocity of a satellite moving about a central body in circular motion

$$v = \sqrt{\frac{G * M_{central}}{R}}$$

where **G** is 6.673 x 10^{-11} N•m^2/kg^2, $\mathbf{M_{central}}$ is the mass of the central body about which the satellite orbits, and **R** is the radius of orbit for the satellite.

The Acceleration Equation

The equation for the acceleration of gravity was given as

$$g = (G \bullet M_{central})/R^2$$

Thus, the acceleration of a satellite in circular motion about some central body is given by the following equation

$$a = \frac{G * M_{central}}{R^2}$$

where **G** is 6.673 x 10^{-11} N•m^2/kg^2, $\mathbf{M_{central}}$ is the mass of the central body about which the satellite orbits, and **R** is the average radius of orbit for the satellite.

Classification of Motions

Orbital Period Equation

The period of a satellite (**T**) and the mean distance from the central body (**R**) are related by the following equation:

$$\frac{T^2}{R^3} = \frac{4 * \pi^2}{G * M_{central}}$$

where **T** is the period of the satellite, **R** is the average radius of orbit for the satellite (distance from center of central planet), and **G** is 6.673 x 10^{-11} N•m^2/kg^2.

There is an important concept evident in all three of these equations - the period, speed and the acceleration of an orbiting satellite are not dependent upon the mass of the satellite.

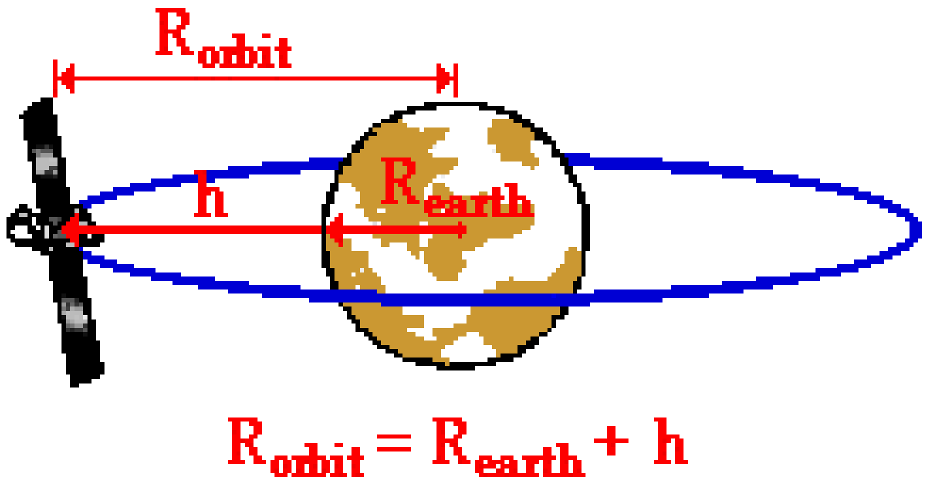
$$v = \sqrt{\frac{G * M_{central}}{R}} \qquad a = \frac{G * M_{central}}{R^2} \qquad \frac{T^2}{R^3} = \frac{4 * \pi^2}{G * M_{central}}$$

Classification of Motions

Practice Problem #1

A satellite wishes to orbit the earth at a height of 100 km (approximately 60 miles) above the surface of the earth. Determine the speed, acceleration and orbital period of the satellite. (Given: Mearth = 5.98 x 10^24 kg, Rearth = 6.37 x 10^6 m)

Given/Known: $R = R_{earth}$ + height = 6.47×10^6 m $M_{earth} = 5.98 \times 10^{24}$ kg $G = 6.673 \times 10^{-11}$ N m^2/kg^2	**Unknown:** v = ??? a = ??? T = ???



Ans: v = 7.85×10^3 m/s, T = 5176 s = 1.44 hrs, and a = 9.53 m/s^2

Classification of Motions

Practice Problem #2

The period of the moon is approximately 27.2 days (2.35 x 106 s). Determine the radius of the moon's orbit and the orbital speed of the moon. (Given: M_{earth} = 5.98 x 10^24 kg, R_{earth} = 6.37 x 10^6 m)

Ans: R = 3.82 x 10^8 m

Practice Problem #3

Knowing that the moon has an orbital period of 27 days at a radius of 3.48×10^8 m from the Earth, determine the orbital radius of a geostationary satellite around Earth (remember geostationary means the satellite has a period of 24 hours).

Ans: R = 1.5 x 10^{11} m

Classification of Motions

2. Circular & Rotational Motion: (ii) Rotational Motion

Drone Motion

Any aircraft or flying machine operated without a human pilot such machines is called an unmanned aerial vehicle (UAV) or Drone. It can be guided autonomously or remotely by a human operator using onboard computers and robots.

During surveillance or military operation, UAVs can be a part of an unmanned aircraft system (UAS), Drones are separately for air and water. Drones have become increasingly popular in recent years. They are used for a variety of purposes, including photography, videography, surveying, inspection, and even delivery. But have you ever wondered how drones work? In this blog post, we'll take a look at the working principle of drones
The basic components of a drone are the frame, motors, propellers, battery, flight controller, and sensors.

Types of drones based on the number of Propellors:
A number of propellors are provided to drones. More propellors improve the stability of drones and load-carrying capacity but such drones need more battery power to drive more motors to get high power. A quadcopter is a mostpopular drone.

- **Bicopter (2 propellers)**
- **Triplecopter (3 propellors)**
- **Quadcopter (4 propellers)**
- **Hexacopter (6 propellers)**
- **Octacopter (8 propellers)**

Classification of Motions

Hexacopter (6 propellers)

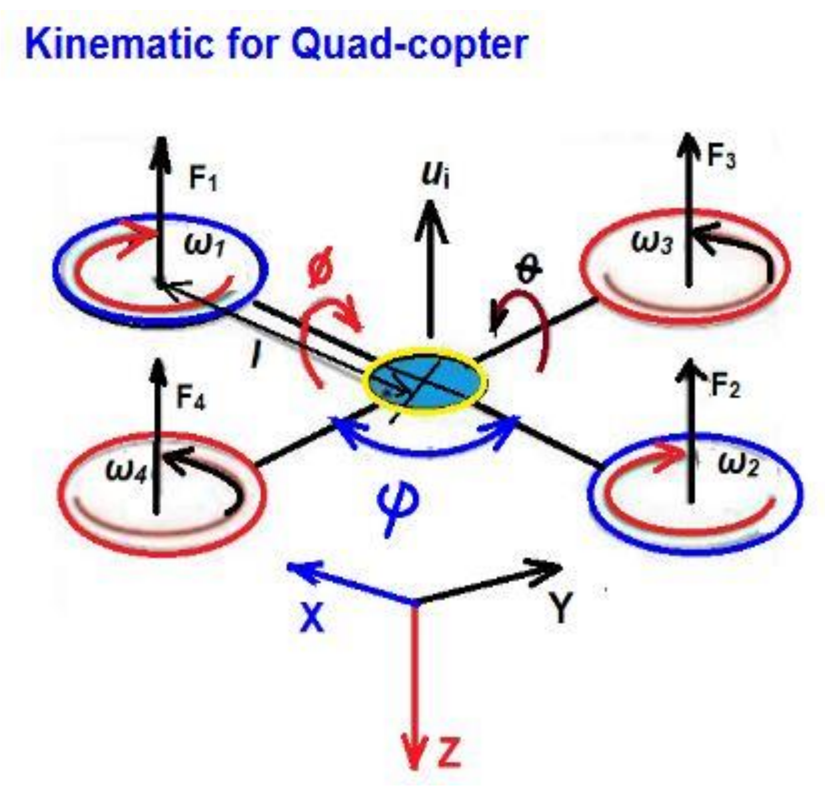
Quadcopter (4 propellers)

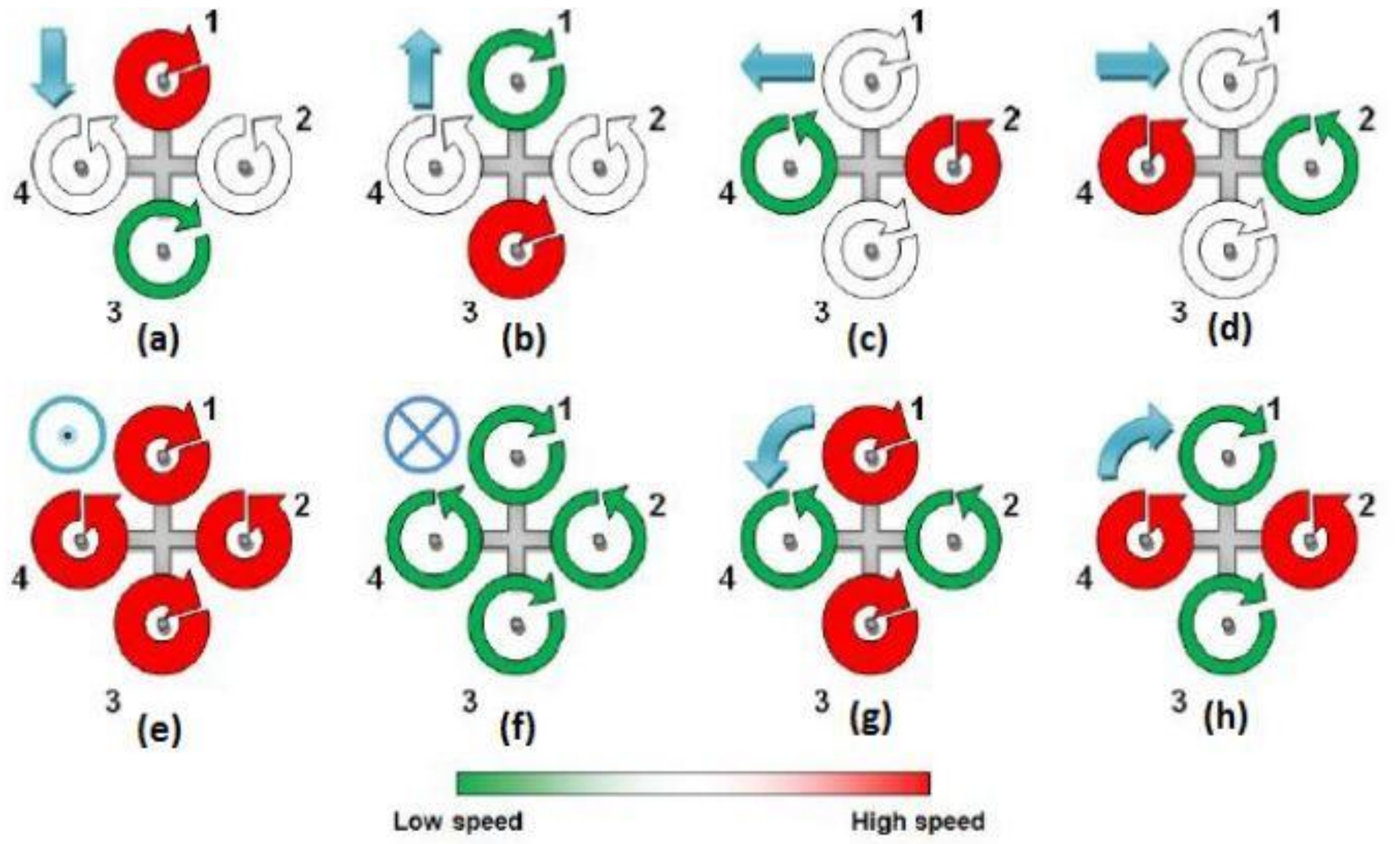
Classification of Motions

Subjects for Drone or UAV: Understanding and development of drones depend on many subjects.

- **Fluid Dynamics or Aerodynamics** **or** Computational Fluid Dynamics (CFD) **:**
  - Fluid dynamics plays an important role in deciding the forces acting on the body of a drone
  - The shape, size, and speed of the propeller and drone depend on the aerodynamics of the propellers or blades
- **Mechanical Design**
  - Rigid body dynamics to study the motion and forces acting on drones
  - Strength of materials
  - Low-weight and rigid materials are selected for drone
- **Electronics and Electrical Components:**
  - An electric motor with and without brush is required to drive the propellors
  - Electronic Speed Controller
  - Flight controller unit and computer processors
- **Radio Communication:** transmitter and receiver for radio signals
- **Battery:** Low weight and high-power wattage battery is important
- **Software-based interface:** data collection and analysis using mobile or computer

Classification of Motions



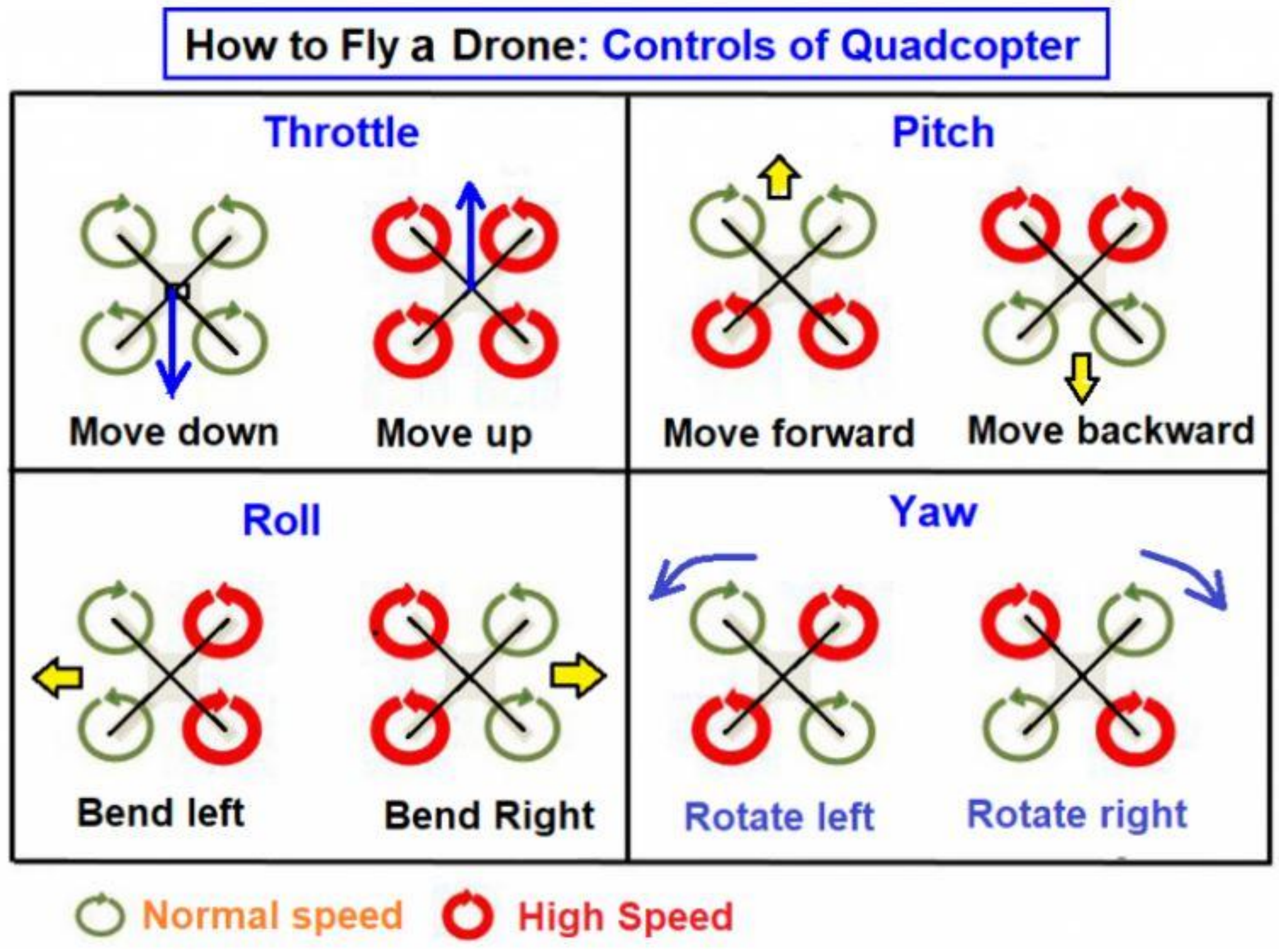


Quadcopter states. (a) Forward motion; (b) backwards motion; (c) movement left; (d) movement right; (e) increase altitude; (f) decrease altitude; (g) leftwards rotation; (h) rightwards rotation.

Classification of Motions

Working Principle of Quadcopter:

- A quadcopter has four propellors at four corners of the frame.
- For each propeller, speed, and direction of rotation are independently controlled for balance and movement of the drone
- In a traditional quadrotor, all four rotors are placed at an equal distance from each other
- To maintain the balance of the system, one pair of rotors rotates in a clockwise direction and the other pair rotates in an anti-clockwise direction
- To move up (hover), all rotors should run at high speed. By changing the speed of rotors, the drone can be moved forward, backward, and side-to-side

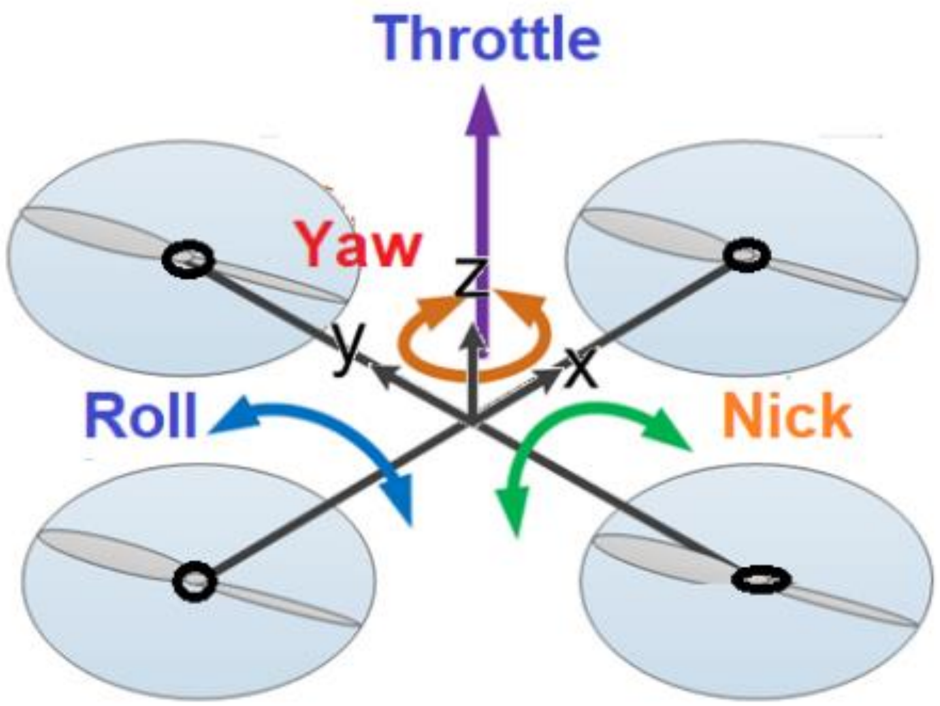


Classification of Motions

Quadcopter Dynamics

- The movement of drones is classified into four types based on the relation motion between four propellors: 1) throttle, 2) Pitch, 3) Roll, and 4) Yaw

The details of quadcopter dynamics are explained in many references



Throttle/ Hover: The up and down movement of the drone is called throttle

If all four propellors run at normal speed, then the drone will move down

If all four propellors run at a higher speed, then the drone will move up. This is called the hovering of a drone

Pitch: movement of a drone about a lateral axis (either forward or backward) is called pitching motion

If two rear propellors run at high speed, then the drone will move in a forwarding direction

If two front propellors run at high speed, then the drone will move in the backward direction

Roll: The movement of a drone about the longitudinal axis is called rolling motion

If two right propellors run at high speed, then the drone will move in the left direction

If two left propellors run at high speed, then the drone will move in the right direction

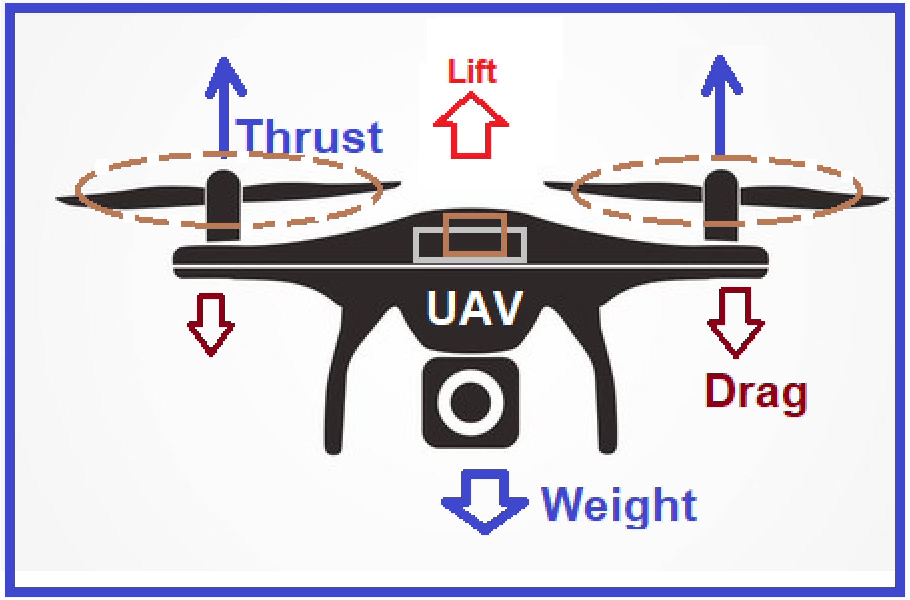
Yawn: the rotation of the head of the drone about the vertical axis (either the left or right) is called Yawning motion

If two propellors of a right diagonal run at high speed, then the drone will rotate in an anti-clockwise direction

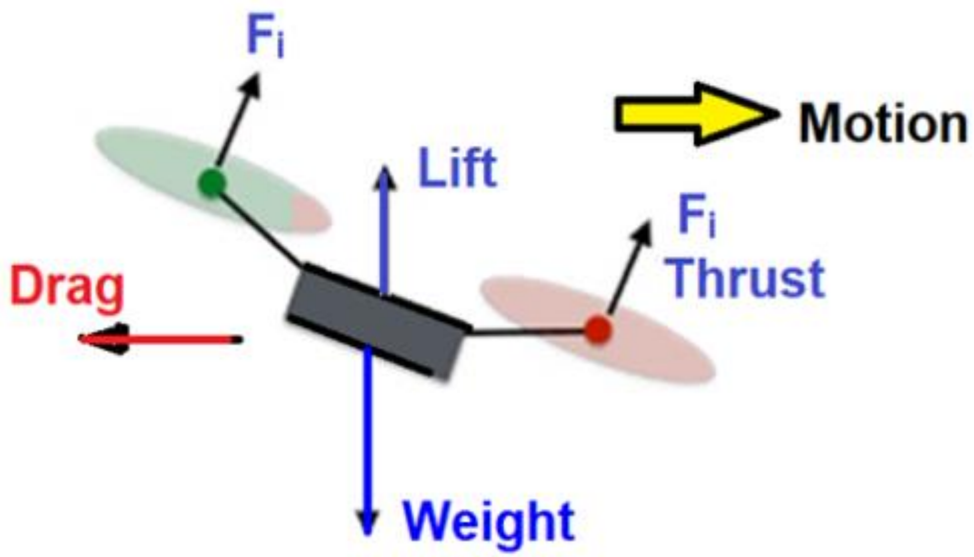
If two propellors of a left diagonal run at high speed, then the drone will rotate in a clockwise direction

Classification of Motions

Working Principle and Components of Drone:



Major forces acting on a Drone



Weight: Due to the mass of the drone, the body mass force always acts in the direction of gravity. The higher the weight of the drone, the more power is required to lift and move the drone. Weight of drone = mass of drone × acceleration due to gravity
Lift: The vertical force acting on the drone is called the lift. This force is due to pressure differences across the drone (in the vertical direction). Hence, the speed, size, and shape of the propeller blade decide the amount of lift force. Lift is essential to lift the body against gravity. To create this force, all four propellors run at high speed to lift the drone
Thrust: The force acting on the drone in the direction of motion is called thrust. However, for drone dynamics, it is normal to the rotor plane. During hovering, the thrust is purely vertical. If thrust is inclined then the drone will tilt forward or backward. This force is essential to move the drone in the desired direction at equal speed. To get the desired motion, two propellors have been given high-speed
Drag: The force acting on the drone in the opposite direction of motion due to air resistance is called drag. This may be because of pressure difference and viscosity of air. To reduce the drag, the aerodynamic shape of the drone is selected.

Classification of Motions

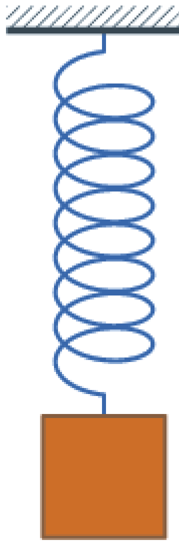
Translational & Rotational Motion analogy: There is a strong difference as well as some similarity between rotational motion and standard translational motion.

Table 3: *The analogies between translational and rotational motion.*

Translational motion		*Rotational motion*	
Displacement	$d\mathbf{r}$	Angular displacement	$d\boldsymbol{\phi}$
Velocity	$\mathbf{v} = d\mathbf{r}/dt$	Angular velocity	$\boldsymbol{\omega} = d\boldsymbol{\phi}/dt$
Acceleration	$\mathbf{a} = d\mathbf{v}/dt$	Angular acceleration	$\boldsymbol{\alpha} = d\boldsymbol{\omega}/dt$
Mass	M	Moment of inertia	$I = \int \rho \lvert\hat{\boldsymbol{\omega}} \times \mathbf{r}\rvert^2 \, dV$
Force	$\mathbf{f} = M\,\mathbf{a}$	Torque	$\boldsymbol{\tau} \equiv \mathbf{r} \times \mathbf{f} = I\,\boldsymbol{\alpha}$
Work	$W = \int \mathbf{f} \cdot d\mathbf{r}$	Work	$W = \int \boldsymbol{\tau} \cdot d\boldsymbol{\phi}$
Power	$P = \mathbf{f} \cdot \mathbf{v}$	Power	$P = \boldsymbol{\tau} \cdot \boldsymbol{\omega}$
Kinetic energy	$K = M\,v^2/2$	Kinetic energy	$K = I\,\omega^2/2$

Classification of Motions

3. *Oscillatory/Vibrational Motion:* When particles are moving **to and fro** or **up and down** towards a mean point, then the motion is said to be Oscillatory/Vibrational Motion. Example: Motion of simple pendulum, Motion of a tuning fork, etc.



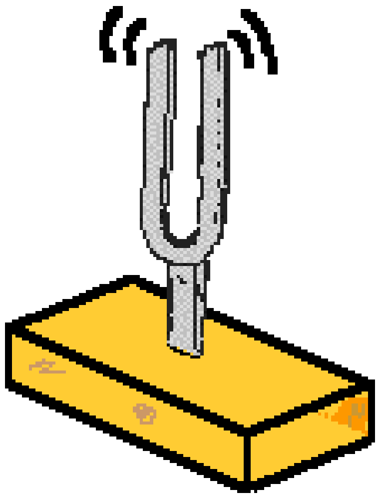


Fig: Tuning fork

R = Rotational factor | Rotational character

V = Vibrational factor | vibrational character

Classification of Motions

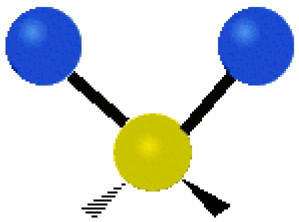
Vibrational factor or Vibrational character:

- ❖ Angular frequency of motion, $\omega = 2\pi f = \frac{2\pi}{T}$
- ❖ Vibration number, $n = \frac{t}{T}$
- ❖ Vibration modes:: Normal Modes of vibrations are: asymmetric, symmetric, wagging, twisting, scissoring, and rocking for polyatomic molecules.
- ❖ Degrees of freedom
- ❖ Vibrational time period, $T = \frac{1}{f} = \frac{2\pi}{\omega}$
- ❖ Vibrational frequency, $f = \frac{1}{T} = \frac{\omega}{2\pi}$
- ❖ Vibrational frequency, $\upsilon = \frac{1}{2\pi}\sqrt{k/_{\mu}}$, where k is force constant and μ *is reduced mass* $\frac{1}{\mu} = \frac{1}{m_A} + \frac{1}{m_B}$
- ❖ Vibrational wavenumber, $\bar{\upsilon} = \frac{1}{2\pi c}\sqrt{k/_{\mu}}$, where *c* is the speed of light
- ❖ Vibrational energy, $E_{\upsilon} = \left(\upsilon + \frac{1}{2}\right) \mathrm{h}\upsilon_0$, υ=0,1,2,…. is vibrational quantum number and $\upsilon_0 = \frac{1}{2}\pi\sqrt{k/_{\mu}}$

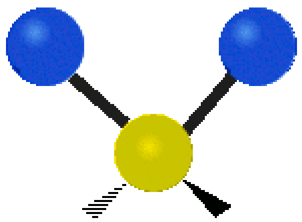
Classification of Motions

 # Vibrational factor depends on Vibrational modes:

Symmetric Stretching



Asymmetric Stretching



Wagging

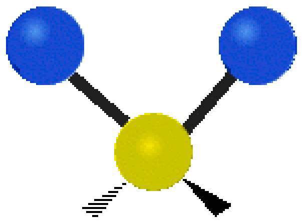
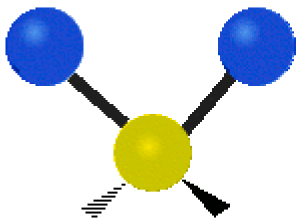
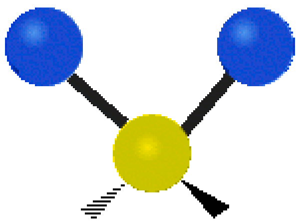


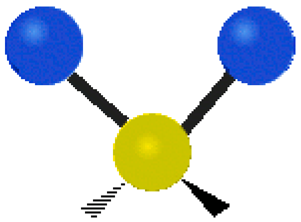
Figure : *Six types of Vibrational Modes. Images used with permission (Public Domain; Tiago Becerra Paolini).*



Twisting



Scissoring



Rocking

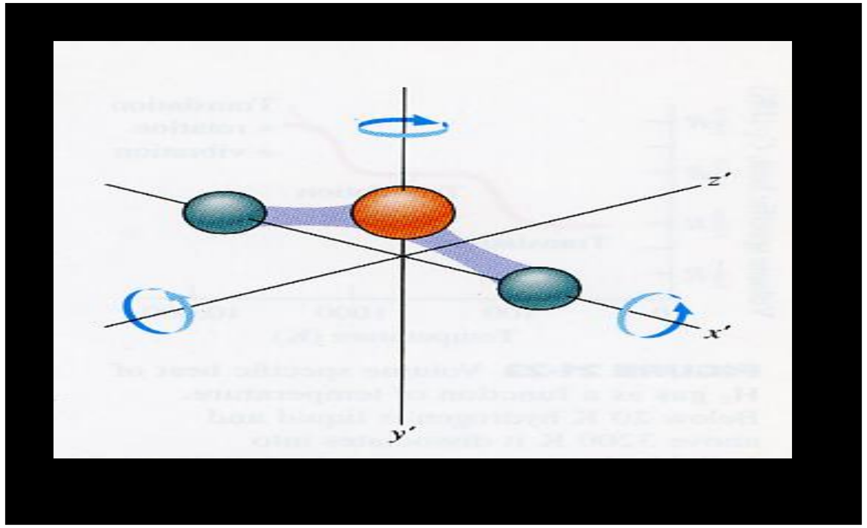
Classification of Motions

Degrees of Freedom

The total number of independent variables required to describe completely the state of motion of a body are called its degrees of freedom.

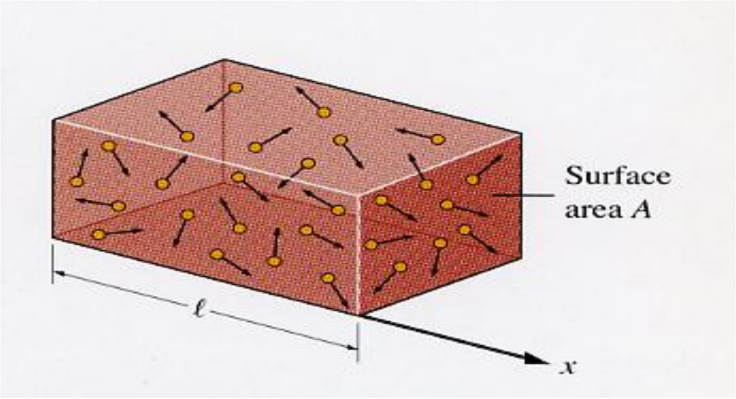
The minimum number of ways in which motion of a body (or a system) can be described completely.

Rotational modes



(internal)

Translational modes



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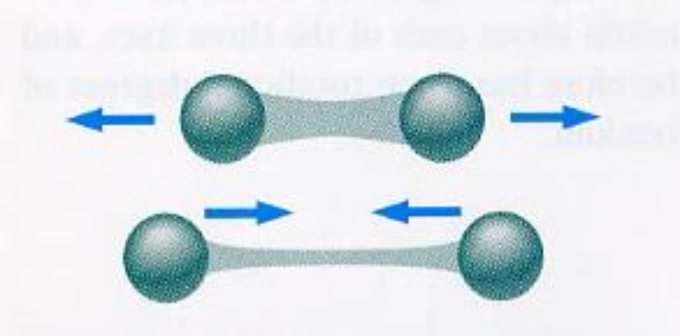
Vibrational modes



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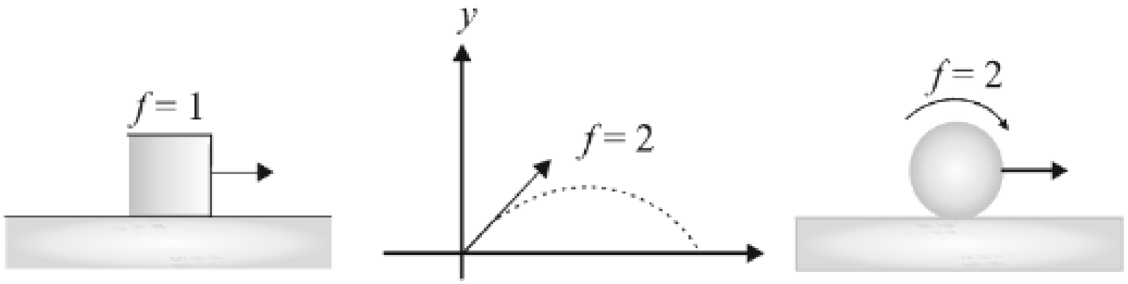
Classification of Motions

Degrees of Freedom Examples

In Fig (a) , block has one degree of freedom, because it is confined to move in a straight line and has only one translational degree of freedom.



(a) (b) (c)

In Fig. (b), the particle has two degrees of freedom because it is confined to move in a plane and so it has two translational degrees of freedom. In Fig. (c), the sphere has two degrees of freedom one rotational and another translational. Similarly, a particle free to move in space will have three translational degrees of freedom.

Degree of Freedom of a Molecules or gas Molecules:

A gas molecule can have, the following types of energies

(i) translational kinetic energy

(ii) rotational kinetic energy

(iii) vibrational energy (potential + kinetic)

Classification of Motions

Degree of Freedom of a Monoatomic Molecule or Monoatomic Gas Molecule:

A monoatomic gas molecule (like He) consists of a single atom. It can have translational motion in any direction in space. Thus, **it has 3 translational degrees of freedom. Degrees of freedom, *f* = 3 (all translational).** It can also rotate but due to its small moment of inertia, rotational kinetic energy is neglected. Example: Ne, He, Ar, etc.

$$\therefore Df = 3\ (3 translational + 0\ rotational)$$ [monoatomic Molecule]

Degree of Freedom of a Diatomic Molecule or Diatomic Gas Molecule:

The molecules of a diatomic gas (like O_2, N_2 and H_2) cannot only move bodily but also rotate about anyone of the three coordinate axes as shown in figure. However, its moment of inertia about the axis joining the two atoms (x-axis) is negligible. Hence, it can have only two rotational degrees of freedom. Thus, **a diatomic molecule has 5 degrees of freedom: 3 translational and 2 rotational.**

$$\therefore Df = 5\ (3\ translational + 2\ rotational)$$ [Diatomic molecule at normal temperature]

Degree of Freedom of a Diatomic Molecule or Diatomic Gas Molecule At high Temperature:

At sufficiently high temperatures, it has vibrational energy as well providing two more degrees of freedom (one vibrational kinetic energy and another vibrational potential energy). Thus, at high temperatures a diatomic molecule has 7 degrees of freedom, 3 translational, 2 rotational and 2 vibrational. Thus,

∴ *Df* = 7 (3 translational + 2 rotational + 2 vibrational) at high temperatures. [Diatomic Molecule at high temperature]

Classification of Motions

Degree of Freedom of a Triatomic Molecule or Triatomic Gas Molecule:

A triatomic molecule, if it is non-linear (H_2S, SO_2, etc.) has 6 degrees of freedom, 3 translational, 3 rotational. Thus,

∴ *Df* = 6 (3 translational + 3 rotational + 0 vibrational) [Triatomic Non-Linear Molecule]

A triatomic molecule, if it is linear (CH_2, CO_2, etc.) has 5 degrees of freedom, 3 translational, 2 rotational. Thus,

∴ *Df* = 5 (3 translational + 2 rotational + 0 vibrational) [Triatomic Linear Molecule]

Degree of Freedom of a Polyatomic Molecule or Linear Polyatomic Gas Molecule:

The molecules of a linear polyatomic gas (like CO_2 and N_2O) cannot only move bodily but also rotate about anyone of the three coordinate axes as shown in figure. Hence, it can have only two rotational degrees of freedom. Thus, **a Linear polyatomic gas molecule has 5 degrees of freedom: 3 translational and 2 rotational.**

$\therefore Df = 5\,(3\,translational + 2\,rotational)$ [Linear Poliatomic molecule]

∴ *Df* = 7 (3 translational + 2 rotational + 2 vibrational) at high temperatures. [Linear Poliatomic Molecule at high temperature]

Degree of Freedom of a Polyatomic Molecule or Non-Linear Polyatomic Gas Molecule:

A non-linear polyatomic molecule (such as NH_3) can rotate about any of three coordinate axes. Hence, it has 6 degrees of freedom 3 translational and 3 rotational. At room temperatures, a polyatomic gas molecule has insignificant vibrational energy.

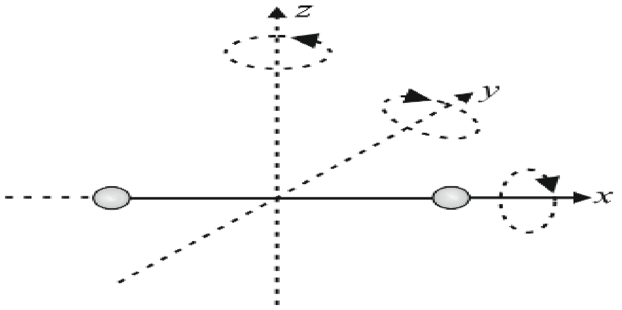
∴ *Df* = 6 (3 translational + 3 rotational + 0 vibrational) [Polyatomic Non-Linear Molecule at Normal Temperature]

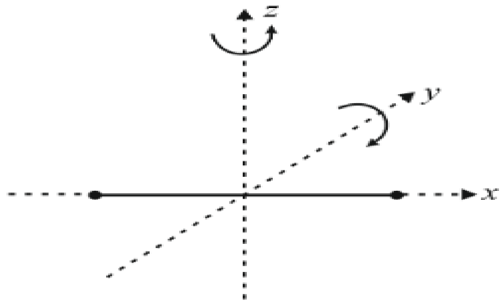
Classification of Motions

Degree of Freedom of a Polyatomic Molecule or Non-Linear Polyatomic Gas Molecule At high Temperature:

But at high enough temperatures it is also significant. So, it has 8 degrees of freedom 3 rotational, 3 translational and 2 vibrational. Thus,

∴ *Df* = 8 (3 translational + 3 rotational + 2 vibrational) [Polyatomic Non-Linear Molecule at High Temperature]

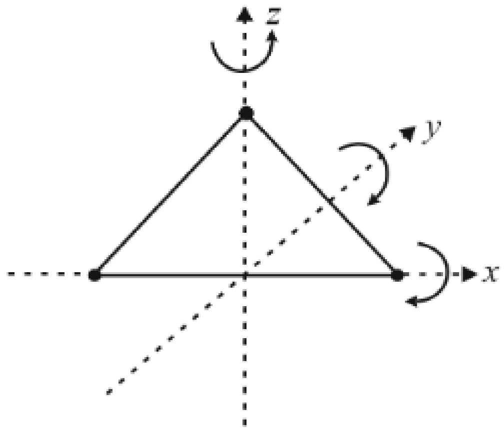




Degree of Freedom of a Solid Molecule :

An atom in a solid has no degrees of freedom for translational and rotational motion. At high temperatures, due to vibration along 3 axes it has 6 degrees of freedom. f = 6 (all vibrational) at high temperatures.

∴ *Df* = 6 (0 translational + 0 rotational + 6 vibrational) [Solid Molecule]

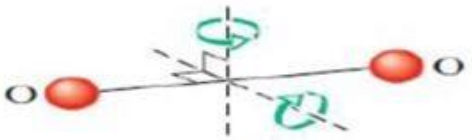


Classification of Motions

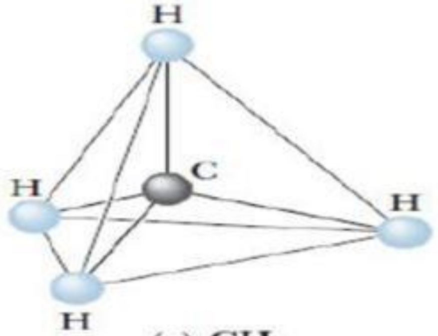
The degrees of freedom of a molecule is the number of independent ways in which the molecule can store energy. Each such degree of freedom has associated with it - on average - an energy of $\frac{1}{2}kT$ per molecule (or, $\frac{1}{2}RT$ per mole).

For a monoatomic molecule can move in three ways along x, y, z . So the degrees of translational freedom is 3. The energy of the molecule is $E = 3(\frac{1}{2}kT) = \frac{3}{2}kT$.
A diatomic molecule can have only two degrees of rotational freedom and rotational energy of $2(\frac{1}{2}kT) = kT$.
For a polyatomic molecule the degrees of freedom is 6 and corresponding energy $6(\frac{1}{2}kT) = 3kT$. Corresponding energy is too law when compared with the experimental data although monoatomic and diatomic molecules agree well.

(a) He



(b) O_2



(c) CH_4

Classification of Motions

The normal modes of vibration are: asymmetric, symmetric, wagging, twisting, scissoring, and rocking for polyatomic molecules

The degrees of vibrational modes for **linear molecules** can be calculated using the formula: $3N-5$

The degrees of freedom for **nonlinear molecules** can be calculated using the formula: $3N-6$

N=n is equal to the number of atoms within the molecule of interest.

Type	Translation	Rotation	Vibration
Linear	3	2	$3n-5$
Non-linear	3	3	$3n-6$

Example-9: How many vibrational modes are there in the CO_2 molecule ?

Soln: $3(3)-5=4$

Example-10: How many vibrational modes are there in the tetrahedral CH_4 molecule ?

Soln: Try yourself 9

Example-11: How many vibrational modes are there in the C_{60} molecule ?

Soln: Try yourself 174

Classification of Motions

3. Oscillatory/Vibrational Motion :

Oscillatory or Vibratory Motion

- Pure Vibratory Motion
- **Reciprocating** Motion

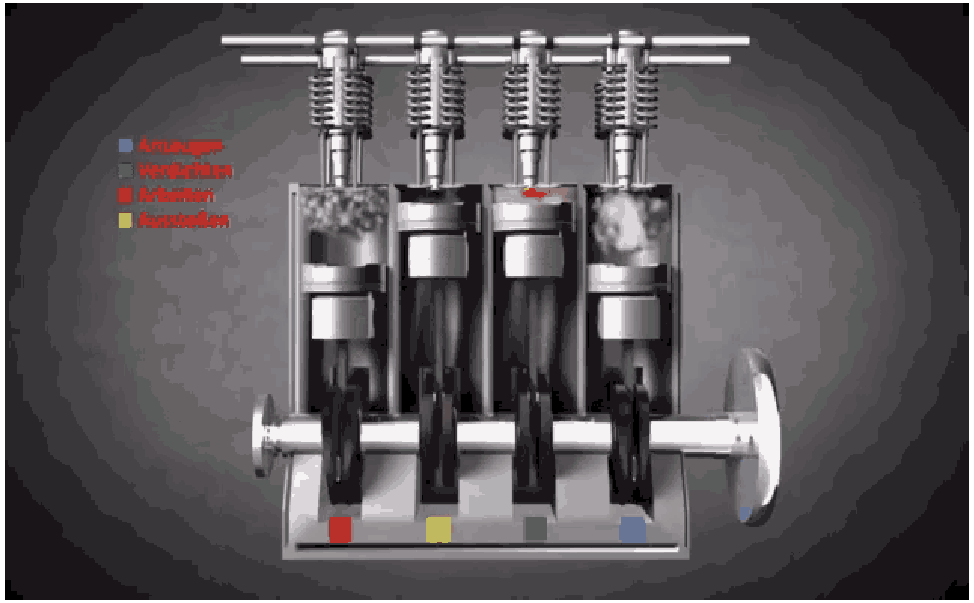


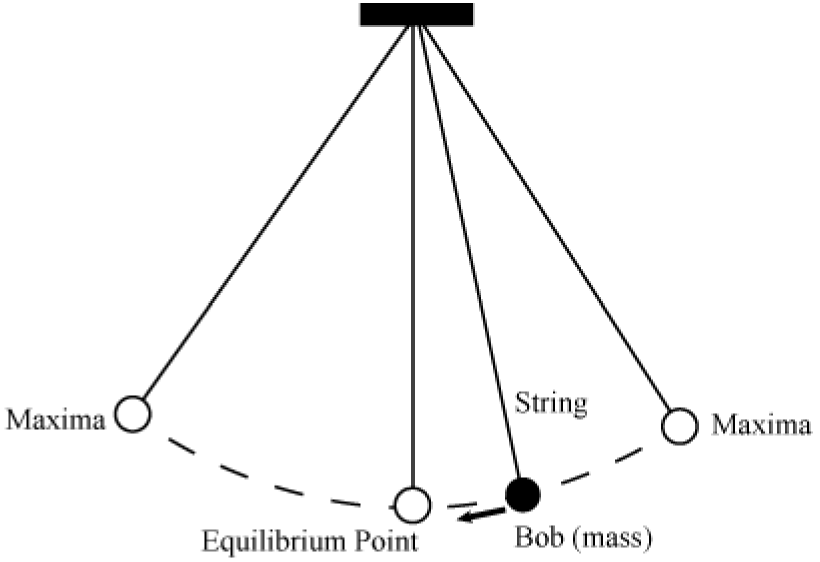
Fig: Pumping Engine motion

Classification of Motions

3. Oscillatory/Vibrational Motion : Oscillatory/Vibratory motion is classified mainly in two branches. (i) Pure Vibratory Motion and (ii) Reciprocating Motion.

(i) Pure Vibratory Motion: When particles are moving **to and fro** or **up and down towards a mean point**, then the motion is said to be pure oscillatory/vibratory Motion.

Example: Motion of simple pendulum, Motion of a tuning fork, etc.



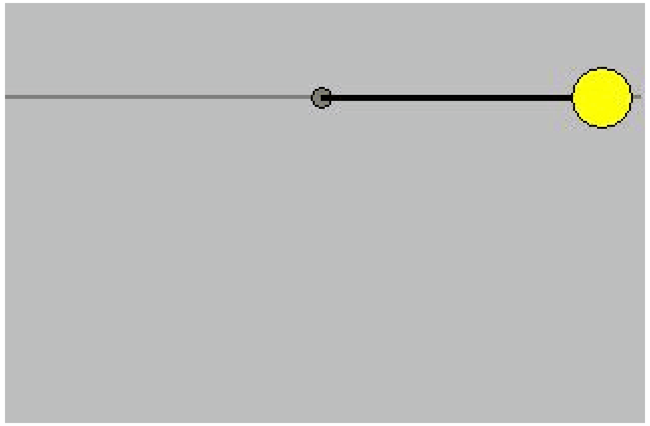


Fig: Motion of a Pendulum

Classification of Motions

3. Oscillatory/Vibrational Motion : Oscillatory/Vibratory motion is classified mainly in two branches. (i) Pure Vibratory Motion and (ii) Reciprocating Motion.

(ii) Reciprocating Motion : *A repetitive and **continuous up and down or back and forth motion in a linear path** is called reciprocating motion.*
It is a repetitive up-and-down or back-and-forth linear motion. The two opposite motions that comprise a single reciprocation cycle are called strokes. Reciprocating motion refers to any repeating motion in a straight line, which includes up and down, side to side, or in and out.
Example: Motion of Pumping Engine, Sewing Machine Needle Motion, etc.

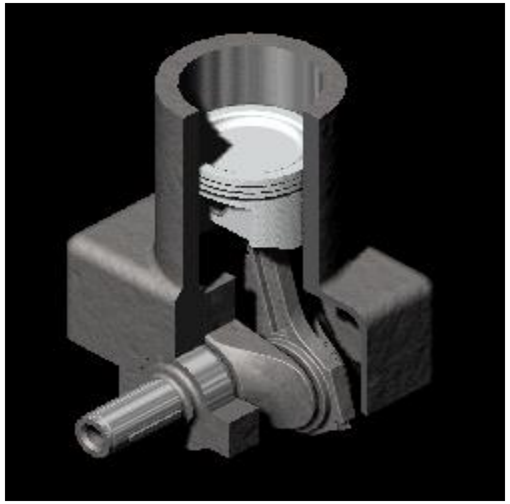


Fig: Shallow water pumping machine

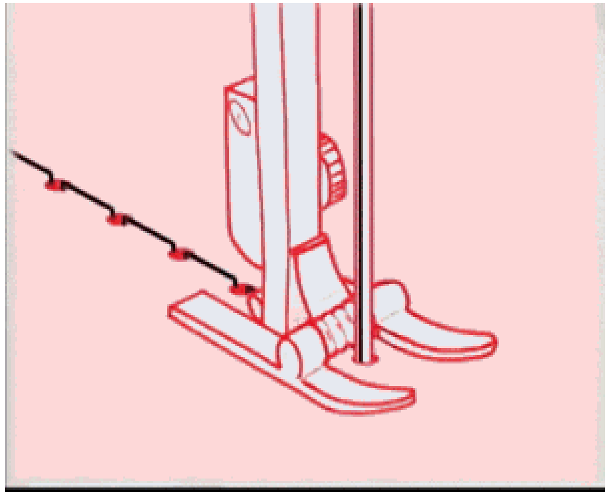


Fig: Sewing Machine Needle

Classification of Motions

4. Rota-Vibratory/Rotational-Vibrational Motion: When the particles are moving under the influence of both rotational and vibrational motion, the overall motion is called Rota-Vibratory/Rotational-Vibrational Motion. Example: Motion of an electric motor, Earth's motion towards the Sun, etc.

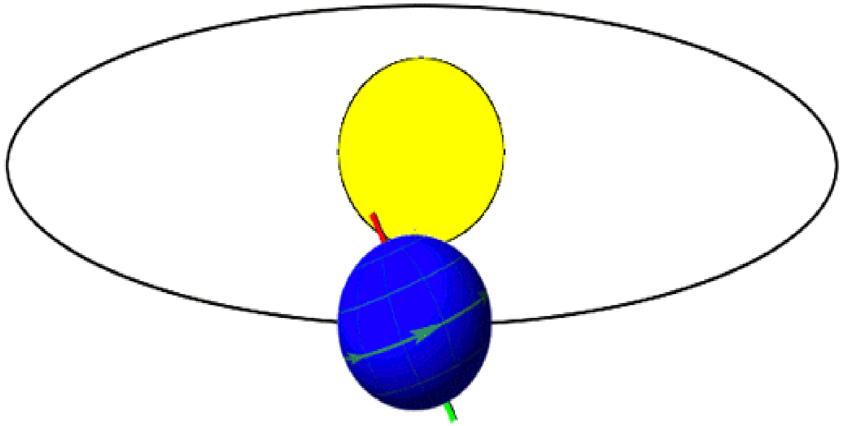
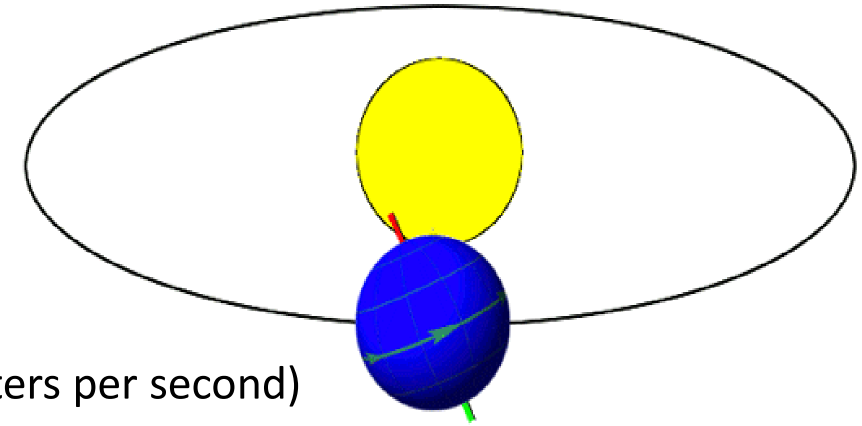


Fig: Earth's rotation towards the sun

Earth orbits around the sun at a speed of 67,100 miles per hour (30 kilometers per second)

Classification of Motions

4. Rota-Vibratory/Rotational-Vibrational Motion:



Earth orbits the Sun at an average distance of 149.60 million km (92.96 million mi), or 8.317 light-minutes

Earth orbits around the sun at a speed of 67,100 miles per hour (30 kilometers per second)

The radius of the earth's orbit around the sun (assumed to be circular) is 1.50 × 10^8 km

One complete orbit takes 365.256 days (1 sidereal year), during which time Earth has traveled 940 million km (584 million mi)

The radius of Earth ranges from a maximum (equatorial radius, denoted a) of nearly 6,378 km (3,963 mi) to a minimum (polar radius, denoted b) of nearly 6,357 km (3,950 mi).

A globally-average value is usually considered to be 6,371 kilometres (3,959 mi) with a 0.3% variability (±10 km) for the following reasons. The International Union of Geodesy and Geophysics (IUGG) provides three reference values: the mean radius (R1) of three radii measured at two equator points and a pole; the authalic radius, which is the radius of a sphere with the same surface area (R2); and the volumetric radius, which is the radius of a sphere having the same volume as the ellipsoid (R3).[2] All three values are about 6,371 kilometres (3,959 mi).

Classification of Motions

Aperiodic Motion:

When particles are moving in indefinite interval of time, then it is called Aperiodic Motion or random motion.

Example: DHM (Damped Harmonic Motion), Tornedo, etc.

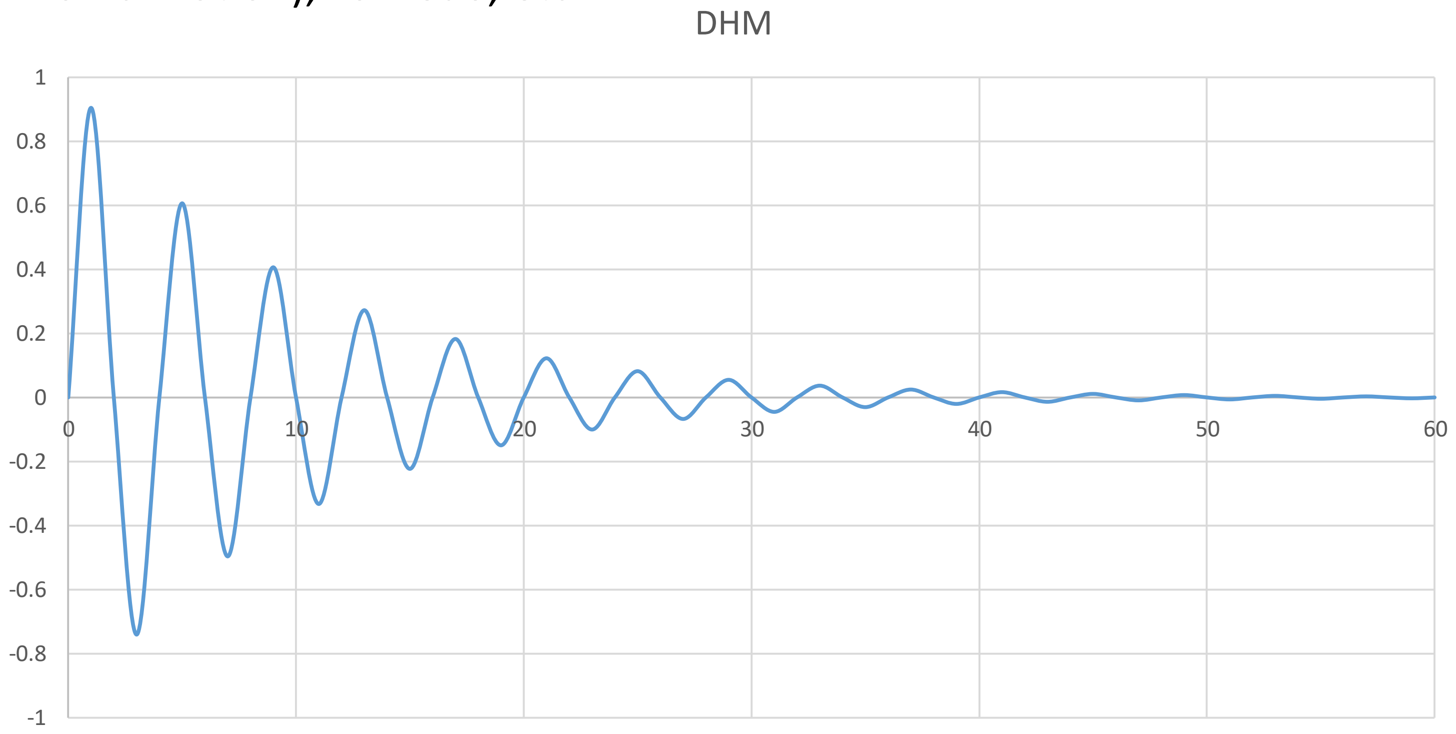
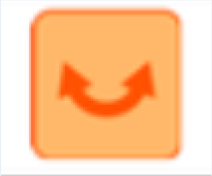


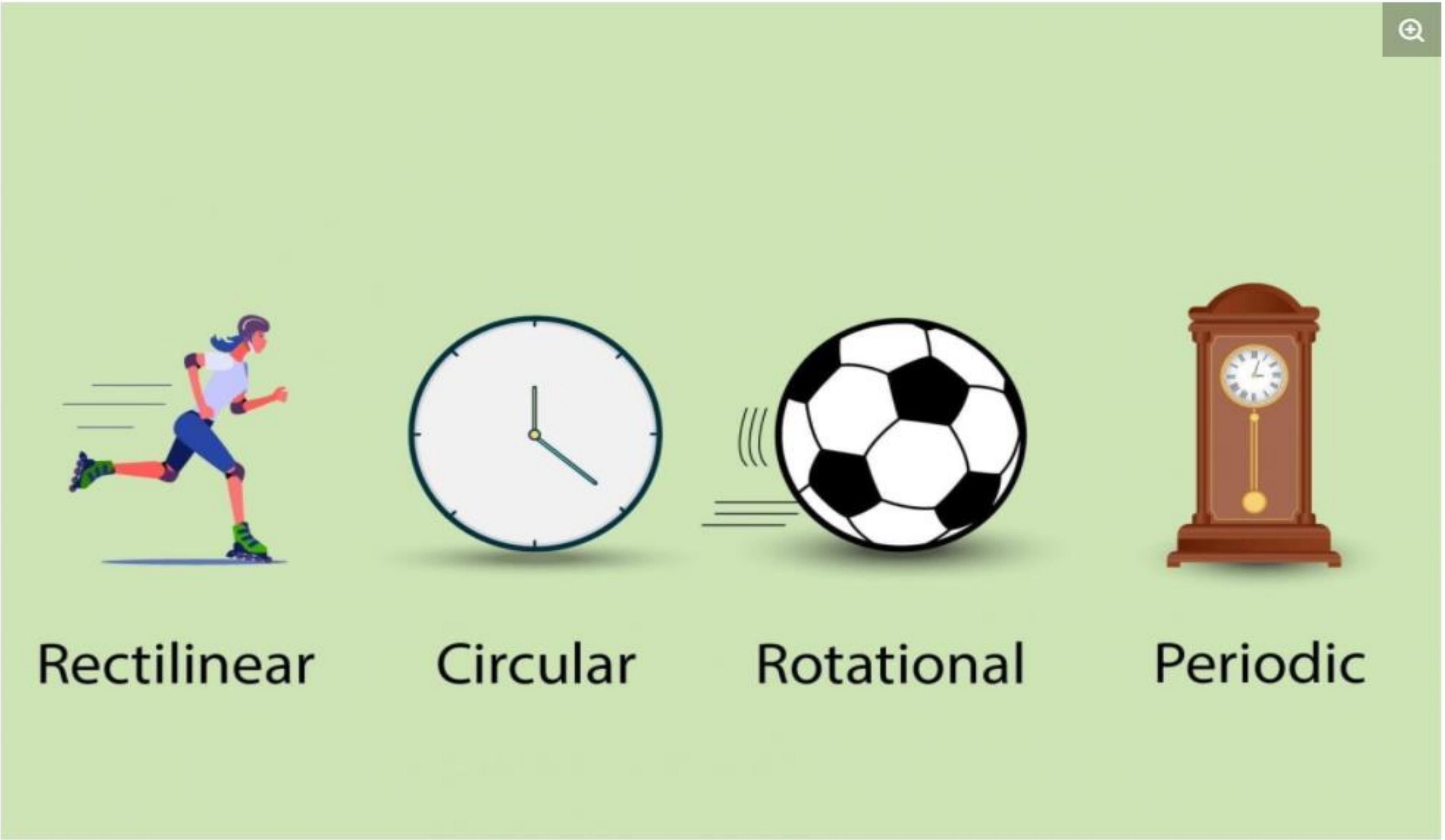
Fig: Aperiodic Wave

Classification of Motions

Motion Image/Think??	Reason	Name of the Motion
	Motion in a straight line indefinitely/definitely	
	Motion in a circle	
	Back and forth motion	
	Oscillation is a back and forth about a pivot point	

Classification of Motions

Now comment about the following motion, which types of it is?????



Classification of Motions

Moreover,

 # Motion can be classified as per the ‘State of the Motion’:

 # Motion can be classified as Per ‘Directions of the Motion’:

 # Motion can be ‘Mixture of Motions/Combination of Motions’:

Classification of Motions

Example-12: In aviation, a "standard turn" for a level flight of a propeller-type plane is one in which the plane makes a complete circular turn in 2.00 minutes. If the speed of the plane is 170 m/s, (i) What is the radius of the circle? (ii) What is the centripetal acceleration of the plane?

Soln:

$$v = \frac{2\pi R}{T} \qquad 170 = \frac{2\pi R}{120} \qquad R = 3247m$$

$$a_c = \frac{v^2}{R} \qquad a_c = \frac{(170)^2}{3247} \qquad a_c = 8.9ms^{-2}$$

Classification of Motions

Example-13: The moon's mass is $7.35x10^{22}$ kg, and it moves around the earth approximately in a circle or radius $3.82x10^{5}$ km. The time required for one revolution is 27.3 days. Calculate the centripetal force that must act on the moon. How does this compare to the gravitational force that the earth exerts on the moon at that same distance? What is the ration of force between them? Mass of the earth is $5.98x10^{24}$ kg.

Soln:

$$F = \frac{m4\pi^2 r}{T^2} = \frac{(7.35\times10^{22}\,kg)(4\pi^2)(3.82\times10^{8}\,m)}{((27.3d)(24hrs)(3600s))^2} = 1.99\times10^{20}\,N$$

$$F = G\frac{mM}{R^2} = \frac{(6.67\times10^{-11}\,m^3kg^{-1}s^{-2})(5.98\times10^{24}\,kg)(7.36\times10^{22}\,kg)}{(3.82\times10^{8}\,m)^2} = 2.01\times10^{20}\,N$$

Classification of Motions

Example-14: A car of mass 1200 kg drives around a curve with a radius of 25.0 m. If the driver maintains a speed of 20.0 km/hr, what is the force of friction between the tires and the road?

Soln: Try yourself $\quad 1481\ N$

Example-15: A 10.0 kg block rests on a frictionless surface and is attached to a vertical peg by a rope. What is the tension in the rope if the block is whirling in a horizontal circle of radius 2.00 m with a linear speed of 20 m/s? What would be the tension in the rope at the top and the bottom of the swing if it were whirled in a vertical circle?

Soln:

$$F_c = ma_c = m\frac{v^2}{R} = (10\ kg)\frac{\left(20\frac{m}{s}\right)^2}{2m} = 2000N$$

$$top:\ T = \frac{mv^2}{R} - mg = 2000\ N - 10(9.8) = 1902 = 1900\ N$$

$$bottom:\ T = \frac{mv^2}{R} + mg = 2000\ N + 10(9.8) = 2098 = 2100\ N$$

Classification of Motions

Example-16: Communications satellites are placed in what are called "geosynchronous" orbits around the earth. This means that the satellite will have an orbital period which matches the period for one earth revolution (one day). (i) At what height above the surface of the earth must one place such a satellite in order to achieve a geosynchronous orbit? How fast will the satellite be moving? Mass of the earth is $5.98x10^{24}$ kg and radius of the earth is 6371 km.

Soln: (i) To geosynchronous, $F_c = F_G$

$$\frac{m_1 v^2}{r} = G\frac{m_1 m_2}{r^2}$$ m_1 cancels and r cancels with r^2

$$v^2 = G\frac{m_2}{r} \rightarrow \left(\frac{2\pi r}{T}\right)^2 = G\frac{m_2}{r} \rightarrow r^3 = \frac{G m_2 T^2}{4\pi^2} \rightarrow$$

$$r = \sqrt[3]{\frac{G m_2 T^2}{4\pi^2}} = \sqrt[3]{\frac{(6.67\times10^{-11})(5.98\times10^{24})(24\times3600)^2}{4\pi^2}} = 42250474.31 \text{ m} = 4.23 \times 10^7 \text{m}$$

Subtract Earth's radius to get height.

$4.23 \times 10^7\text{m} - 6.371 \times 10^6 = 3.58 \times 10^7$ m

Classification of Motions

(ii) How fast will the satellite be moving?

$$v = \frac{2\pi r}{T} = \frac{2\pi(4.23\times10^7)}{(24\times3600)} = 3076\frac{m}{s}$$

Example-17: A fly of mass 2.00 g is sunning itself on a phonograph turntable at a location that is 4.00 cm from the center. When the turntable is turned on and rotates at 45.0 rev/min, calculate the centripetal force needed to keep the fly from slipping?

Soln: Try yourself 0.002 N

Example-18: A 35.0 kg boy is swinging on a rope 7.00 m long. He passes through the lowest position with a speed of 3.00 m/s. What is the tension on the rope at that moment?

Soln: Try yourself 388 N

Example-19: An athlete twirls a 8.0 kg hammer around his head at a rate of 2.00 rev/s. If his arms are 1.00 m long, compute the tension in them.

Soln: Try yourself 1263 N

Example-20: The radius of the earth is $6.37x10^6$ m. How fast, in m/s, is a tree at the equator moving because of the earth's rotation?

Soln: Try yourself 463 m/s

Classification of Motions

Example-21: Bob starts running around from the northernmost point of a circular track. He takes 1.93 minutes to complete each lap of the track. The track has a radius of 200 m. He runs clockwise. Where is Bob relative to his starting point after running for one hour?

(x,y)=(105.159, 170.1220181)

200 m – 170.122 m=29.87798 m

Soln: Try yourself $\theta = \theta_0 - \omega t$=?

Direction=105.159 m East, 29.87798 m south of his starting point

Use radian mode $(rcos\theta, rsin\theta)$ =?

By the distance formula or Line Distance between two points = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ =???

Example-22: Maria starts at the southernmost point of a circular track and run clockwise at 4 meters per second. The track has a radius of 50 m. After running for 10 seconds, how far is she from her starting point? After 1000 seconds?

50 m – 34.8354 m=15.1646 m

Soln: Try yourself

$\theta = \theta_0 - \omega t$=?

$(rcos\theta, rsin\theta)$ =?

(x,y)=(-35.8678, -34.8354)

Direction= 35.8678 m West, 15.1646 m North of his starting point

Line Distance between two points = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ =38.9418 m

and

(x,y)=(49.6944, 5.5195)

Direction=?

Line Distance between two points = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ =74.5113 m

Example-23: If a roller coaster is successfully executed a 20 m radius vertical loop, what speed must each car on a roller coaster be travelling at the top of the loop? Why are there no minimum or maximum weight restrictions?

Soln: Try yourself 14 m/s

Classification of Motions

Thank You

The END